

HOLISTIC HEALTH &  
**PRODUCT  
PRINCIPLES**

UNDERSTANDING THE BASICS  
OF NATURAL MEDICINE

FOUNDATION  
3<sup>rd</sup> EDITION

# INTRODUCTION

## Welcome to the Principles of Health!

This course will introduce you to the concept of food as medicine, it will teach you to use lifestyle and supplementation to manage stress and prevent imbalance, and it will expose you to the healing world of plant medicines and homeopathy.

Did you know that most chronic diseases are preventable?

This means that what we eat, how we manage stress, how well we sleep and rest, whether we exercise, and how we digest our emotions and life experiences will all determine if we are likely to fall ill with a chronic disease such as cancer, diabetes, high blood pressure or heart disease.

Of these, nutrition and lifestyle play the biggest roles - they are extremely powerful tools that we can safely use to prevent disease from occurring in the first place, or reversing illness when the body is out of balance.

Feeling vibrantly alive and well is our natural state of being. When we are not at ease on a physical, spiritual or mental level, we need deep nutrition from clean food sources; gentle, non-invasive holistic therapies; and trusted botanicals and nutraceutical products to support our body's innate healing wisdom.

It is within our power to move towards wellness with small changes in our every day life. It is as easy as taking the first step. Let's get started.



# IMPORTANT TERMS

## **Holistic Health**

Looks at an individual from a whole perspective and acknowledges the mind, body, spirit connection and the importance of balancing all aspects of one's life in order to be healthy. Our emotions, beliefs, values and attitudes affect our health and must be part of the cure.

## **Natural Medicine**

Any system of medicine that complements and enhances the body's natural capacity to heal by restoring balance without the use of synthetic drugs or chemicals. There is an inherent self-healing process or life force in us that is ordered and intelligent.

By using methods and medicines which minimise the risk of harmful side effects and using the least force necessary to diagnose and treat a person, this self-healing process is supported to do its healing work.

## **Acute vs Chronic Conditions**

An acute illness is usually of short duration and of recent onset and symptoms may be severe. Examples are the flu and the cold.

A chronic condition is a disease that is persistent or long lasting in its effects. The term is usually applied when the illness lasts for longer than 3 months. Arthritis, asthma, cancer, diabetes and HIV/AIDS are examples.

## **Symptomatic treatment vs Preventative measures**

Symptomatic treatment: treating or suppressing the symptoms of the disease and not considering the causes.

Preventative measures: Using nutrition and a healthy lifestyle to prevent disease from occurring in the first place. Seeking to identify and remove the underlying causes of the illness rather than to merely suppress the symptoms.

## **Contra-indication**

A condition or factor that serves as a reason to withhold a certain medical treatment or medicine due to the harm that it could cause the patient.

# 1

## MODULE ONE

### NUTRITION

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*"Conventional medicine is by far the best for acute illness. But food is the best medicine for chronic disease. It works faster and better, and is cheaper than medication. And all the side effects are good ones. No drug can do what (real) food can do."*

**- Dr Mark Hyman, Functional Medicine Institute**

# 1. A HEALTHY DIET

*"Good nutrition will prevent 95% of all disease" – Linus Pauling*

## Guidelines for a healthy and balanced diet

Diet means "Way of Life" – a way of eating that you can follow for the rest of your life:

- Should include all 3 macronutrients: healthy carbohydrates, fats and proteins
- Should include micro-nutrients (vitamins and minerals):  
7 to 10 cups of fruits and vegetables per day
- Food should be fresh, whole and unprocessed – home-made is always best
- Eating a wide variety and many colourful foods is important
- Include some raw food in the diet every day
- Eat local and seasonal as far as possible
- Listen to your body – nutrition should be individualised



## HEALTHY CARBOHYDRATES

- Carbohydrates are broken down into glucose which is used by the body for quick energy.
- The right kinds of carbohydrates provide a slow steady supply of energy and are packed with nutrients and fibre.
- They have a low Glycemic Load or Index (GL/GI) meaning they get released into the blood stream slowly so they help to maintain balanced blood sugar levels.
- Because they have not been refined they may take longer to cook.

### Healthy carbohydrates that can be found at Wellness:

- Grains: brown rice, oats, buckwheat, quinoa, rye, spelt, barley, whole wheat flour, teff, sorghum and millet.
- Vegetables
- Fruit: fresh and dried
- Dried Legumes: beans, lentils and chickpeas



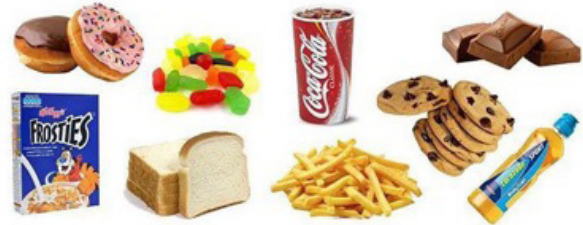
### Unhealthy Carbohydrates:

- Have been refined and stripped of fibre and nutrients
- These carbs cause a spike in blood sugar:
  - Anything made with white or refined flour
  - Simple sugars such as fructose, lactose, maltose, mannitol, dextrose, sucrose, corn syrup, fruit concentrate
  - Carbonated drinks & fruit juices
  - Most processed and fast foods

#### ✓ COMPLEX CARBS



#### ✗ SIMPLE CARBS



## HEALTHY FATS

### Fats are essential for maintaining a healthy body:

- Fats are a clean burning and long-lasting source of energy in the body
- Every cell membrane is made from fat
- The fat-soluble vitamins A, D, E and K are absorbed in the presence of fat
- Many hormones are made from cholesterol
- Fats are important for brain health

The body needs a wide variety of fats and the best forms are those found in *nature*.

### SATURATED FATS

Saturated fats may come from animal or plant sources. They are solid at room temperature. As they tend to raise cholesterol limit their intake and balance with unsaturated fats.

#### Saturated animal fats include:

- Lean meats from pasture raised/grass fed animals
- Full fat dairy products and eggs from pasture raised/grass fed animals

#### Saturated plant fats include tropical oils:

- Coconut oil
- Palm kernel oil

**Saturated fats good to cook with in small amounts:** coconut oil and ghee (do not get damaged when exposed to heat).



## UNSATURATED FATS

**Unsaturated fats** come from plant oils or fatty fish and are a healthy option as they contain essential omega 3 and 6. The only ones to use should be unrefined, cold pressed or extra-virgin, this means they have not been heated and all the health benefits are still available. Store in a cool dark place. These fats are liquid at room temperature. They tend to lower cholesterol.

### Healthy plant and fish oils available at Wellness:

- Olive oil
- Avocado oil
- Hemp seed oil
- Flaxseed oil
- Sesame seed oil
- Omega 3 oils from fatty fish

### Healthy food sources include:

- Avo's
- Olives and olive oil
- Nuts and seeds
- Fatty fish



**Unsaturated fats good to cook with in small amounts:** Avo oil, which has a high burning point.

Avoid low fat products as sugar is often added to boost taste.

### Essential fatty acids:

- The unsaturated fatty acids Omega 3 and Omega 6 need to be taken in from the diet as they cannot be made by the body. Omega 3 is anti-inflammatory while Omega 6 tends to be inflammatory in the body when it is heated in the manufacturing and cooking process.
- The ratio of these two is important and generally we consume too much Omega 6 from processed and junk foods. It is thus important to recommend increasing the intake of Omega 3 rich foods or to take it as a supplement.

### Good sources of Omega 3 include:

- Oily fish: sardines, pilchards, mackerel, snoek, tuna and salmon. The smaller fish are better as they are less likely to be contaminated with heavy metals
- Walnuts and seeds, especially flaxseed/linseed, chia seeds, pumpkinseeds and hemp seeds
- Green leafy vegetables and algae

### Unhealthy Fats:

- Trans fats/ hydrogenated or partially hydrogenated oils – are very processed, heated and found in most packaged and junk foods, as well as margarine
- Heated oils or fats – heating changes the molecular structure of oils and fats



## HEALTHY PROTEINS

The body needs protein to grow and repair - the building blocks of proteins are **amino acids**. We need to consume about 0.8 g of protein per kg of body weight per day to stay healthy. This amounts to 56 grams per day for the average sedentary man and 46 grams per day for the average sedentary woman. If we are very active or want to increase muscle mass, the amount rises to between 1.2 and 1.7 grams of protein per kilogram of body weight per day.

Protein can come from animal or plant sources.

**Animal protein** should come from clean sources - free range, pasture raised and grass fed animals who have not been fed genetically modified grains and corn, and have not been given antibiotics or growth hormones.

Animal products contain enough of every one of the 9 essential amino acids and are considered complete proteins.

### Sources of animal protein:

- Fish, beef, lamb, pork, chicken, duck and game
- Eggs and dairy products from cows or goats
- Sustainably and wild caught fish

**Plant proteins** are an important part of a healthy diet and are especially important for vegetarians and vegans. Plants are usually too low in or missing one or more of the 9 essential amino acids. They're considered incomplete protein sources. It is still possible to get enough of each essential amino acid by eating a varied diet and combining complementary plant proteins throughout the day. Combining grains and legumes will provide complete proteins, for example beans on toast or hummus and pita bread. Sprouting increases amino acids in grains, for example in Ezekiel bread.

### Plant foods high in proteins:

- Superfoods such as spirulina, chia seeds and hemp seeds
- Pseudocereals such as quinoa, buckwheat and amaranth
- Legumes such as beans, chickpeas and lentils
- Nuts and seeds



## HEALTHY FIBRE

### A healthy diet contains all the right kinds of fibre:

• **Insoluble fibre** – this is the indigestible part of fruit and vegetables, usually the skin or leaves, and is often referred to as roughage. This makes up the bulk of stools and has a laxative effect helping to move the stool through the colon. Too much insoluble fibre may be abrasive and cause loose stools.

• **Soluble fibre** – this is the digestible fibres that has the capacity to absorb water. It allows a well-formed soft stool to be formed. Soluble fibre can be found in the following foods: chia, linseeds/flaxseeds, oats, barley, rice, apples, pulses (beans & lentils), fruit and vegetables. Psyllium husk powder is also a good way to supplement soluble fibre (increase water intake).

• **Prebiotic fibre** – this form of soluble fibre is beneficial for the gut flora (microbiome). It helps to increase the naturally occurring probiotics in the gut. The two well-known forms are fructooligosaccharide (FOS) and inulin. They are found in the following foods: Baobab powder, Jerusalem artichokes, artichokes, onions, garlic, leeks, chicory, bananas, papaya, mango and asparagus.



## HEALTHY WATER

We cannot survive without water - the average body is 70% water and it is essential for almost every bodily process, including digestion, absorption, circulation and excretion. Water is used to transport nutrients, it maintains the body temperature and is also key for regulating pH. Water is needed to flush toxins out of the body, without adequate water toxins are not eliminated and can build up. Drinking adequate water can also slow the ageing process.

It is important to avoid becoming dehydrated as this can cause many health issues including headaches, poor brain function, bowel and bladder problems.

### Good Quality Water:

- **Tap water** contains chlorine in small amounts to kill disease causing bacteria. It is important to use a water filter for tap water to remove the chlorine. Carbon is a good way to filter the chlorine out of tap water.
- Buying **bottled mineral water** means that the water has been tested for the minerals it contains. But because it is often stored in plastic the chemicals found in plastic and plastic molecules can leach out into the water. Plastic water bottles should not be left in the sun as this causes further leaching.
- **Ground water and spring water** need to be tested to make sure they are safe for drinking. Spring water is ground water that is forced up out of the ground.
- **Filtering water** is a good option so that you have control of the quality of water you consume.
- Adding minerals and electrolytes to drinking water improves its absorption. Add a pinch of Himalayan salt or a squeeze of lemon juice.



## BENEFICIAL FOODS TO ADD TO THE DIET

### Fermented Foods

People have long recognised the benefits of fermented foods. These cultured or naturally fermented foods have increased probiotic content and higher nutritional value. They can aid in maintaining a healthy gut which is important for good absorption, healthy immune function and brain health.

- **Fermented foods:** sauerkraut, kimchi, miso, tamari, tempeh, cottage cheese, plain and coconut yoghurt, cultured cream, cultured butter, apple cider vinegar
- **Fermented drinks:** kefir milk and water, kombucha, cultured buttermilk

### Bone Broth

Made from slow cooking free range bones with a little vinegar which helps to release all the nutrients in the bones including minerals, collagen and gelatine. This helps heal the digestive track, supports the immune system and is also beneficial for joints and the skin.

At Wellness Warehouse we sell bone broth as frozen liquid stock, or in powdered form.



## FOODS TO AVOID

### Food Additives:

- Food that has been processed and contains additives such as preservatives, colourants, artificial flavourings and flavour enhancers such as MSG.



### Wellness Warehouse No-No's:

Wellness has a list of unacceptable ingredients. When sourcing and evaluating products, the merchandising team are very strict about the ingredient list. WE CHECK THE LABEL SO YOU DON'T HAVE TO. Extensive research goes into ensuring that all the foods on the Wellness shelves are free from the following three ingredients:

#### • Battery eggs

The vast majority of egg-laying hens in South Africa are kept in very small cages or crowded conditions. Caged hens also suffer from the denial of many natural behaviours such as nesting, perching and dustbathing, all important for hen welfare. Wellness doesn't allow any battery eggs, or battery chickens to reach the shelves.

#### • Bleached flour

White flour is the white powder that is left after stripping all the nutrients and fibre out of a grain of wheat, and then chemically bleaching it. White flour is then dried in kilns at high temperatures which kill of any remaining nutrients. When this flour is so stripped of goodness, the shelf life becomes increasingly longer. It is a good material to mix with sugars to make breads and baked goods with a long shelf life. Wellness chooses wholegrains instead of bleached white flour.

#### • Non-GMO

Genetically modified organisms (GMOs) are living organisms whose genetic material has been artificially manipulated in a laboratory through genetic engineering. Certain crops have had their genetic code altered to make them resistant to a particular pesticide or herbicide, for instance. This allows more of the pesticide or herbicide to be used on the field without damaging the crop. There are concerns around the safety of GM crops for human consumption as well as their impact on the environment. GMOs are linked to health problems and environmental damage.

The most commonly grown GM crops in South Africa are soya, wheat and corn. Wellness says NO to GMO!

All organic food is by nature GMO-free.

## FOODS IN EXCESS

### Sugar:

One of the main reasons for the many chronic health conditions experienced today is because of the over consumption of refined sugar and high fructose corn syrup. These non-foods cause inflammation in the body, which is the underlying cause of many illnesses. Raw sugar in its natural state is full of healthy nutrients, but when sugar is processed the nutrients are removed. Molasses is all the goodness that is left behind when making refined sugar.

**Natural sugar options available at Wellness include:** honey, coconut sugar or nectar, raw sugar, agave, stevia, maple syrup, date syrup and molasses. Mesquite, Lucuma and White Mulberries are naturally sweet superfoods that are good to add to smoothies. It is still advisable to consume as little as possible of these sweet foods. Use in small amounts for baking.

Natural sweeteners that will not affect blood sugar levels **suitable for diabetics:** monk fruit, xylitol, erythritol and stevia.

**Avoid artificial sweeteners** such as Aspartame, Acesulfame-K and Sucralose as they may cause cancer. They are also called “non-nutritive sweeteners” on labels. Avoid foods and drinks that have “Zero” or “Lite” in the name as they probably contain artificial sweeteners.

### Salt:

Table salt is processed and only contains sodium, too much sodium can cause water retention and is one of the main reasons for high blood pressure. Natural salt is full of minerals and trace minerals and is a healthier option.

**Healthy salt options at Wellness include:** Himalayan salt, Khoisan salt, Maldon salt.

Salt mixed with herbs and spices: Herbamare and Trocomare, and grinders with salt and herbs/spice mixes.

We do need to consume small amounts of healthy salts to ensure good electrolyte levels and to enable our cells to communicate.



# QUESTIONS:

1. Give 4 basic guidelines for a healthy and balanced diet
2. Why are refined carbohydrates bad for your health?
3. Name 3 foods that are regarded as unhealthy carbohydrates
4. Name 3 healthy grains that you can find at Wellness Warehouse
5. Give 4 reasons why it is important to consume healthy fats
6. Name 3 healthy saturated fats
7. Name 3 healthy unsaturated fats
8. Name the 3 fats that are advisable to cook with
9. Why are they good to cook with?
10. What does it mean when we say animal protein should come from clean sources?
11. What are 2 good examples of plant proteins?
12. How would you explain soluble fibre to a customer and what food choices may you want to suggest?
13. Why is prebiotic fibre important?
14. Why is it important to consume fermented foods and drinks?
15. List 6 fermented foods or drinks that we sell at Wellness Warehouse?
16. Name 3 Wellness Warehouse no-no's
17. Name 4 healthier sugar replacements
18. Name 3 sugar replacements you could recommend to someone who is diabetic
19. Why is it not advisable to replace sugar with artificial sweeteners?
20. Name 3 healthy sodium replacements
21. Is a salt-free diet advisable?

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## 2. FOOD ALLERGIES & SENSITIVITY/ INTOLERANCE

### FOOD ALLERGIES

A food allergy is an **immune-mediated** reaction to certain foods. Your immune system is your body's defense against invaders like bacteria, fungus, or the common cold virus.

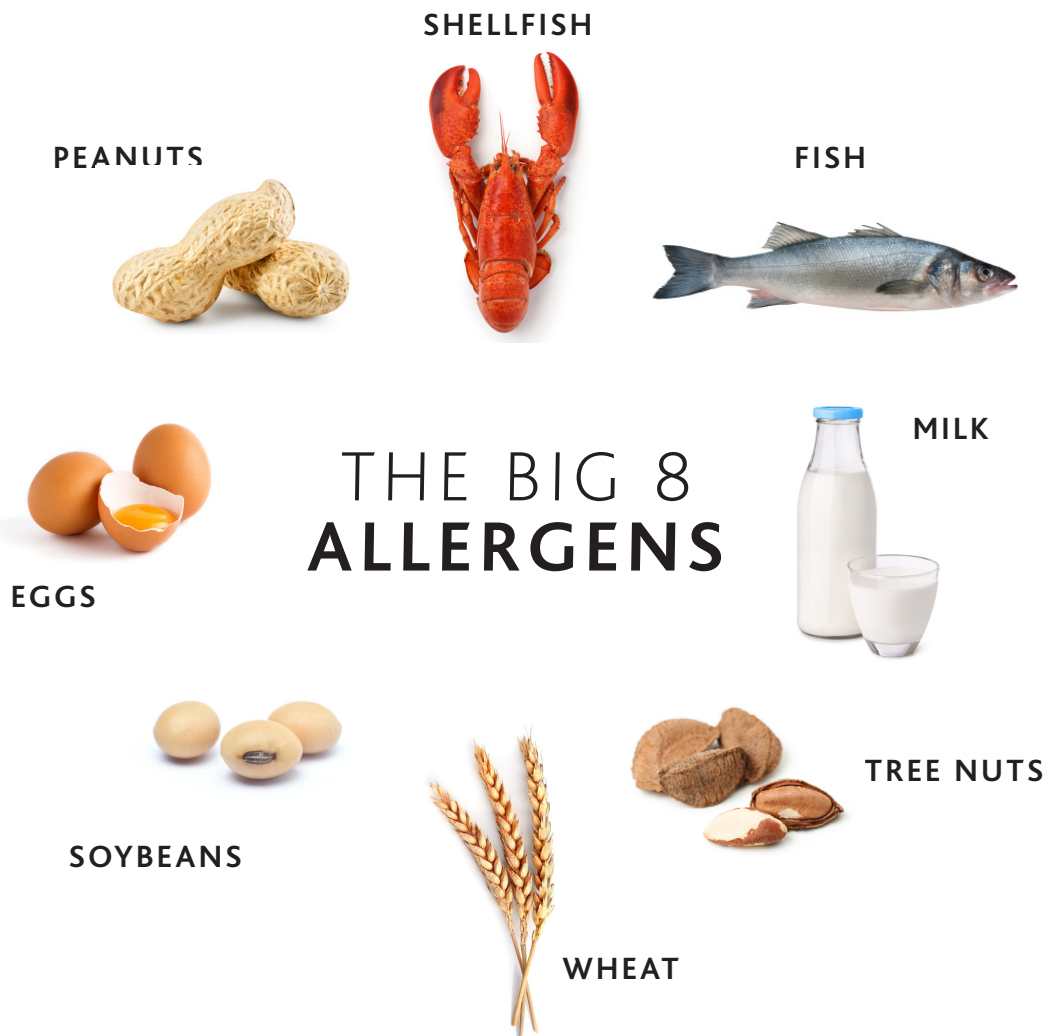
You have a food allergy when your immune system identifies a protein in what you eat as an invader and reacts by producing antibodies to fight it.

Food allergies can be fatal. In extreme cases, ingesting or even touching a small amount of the allergen can cause a severe reaction. The symptoms of an allergy tend to come on quickly after eating a particular food.

#### Symptoms of food allergy include:

- Acute skin reactions, like hives, swelling, and itching
- Anaphylaxis, including difficulty breathing, wheezing, dizziness, and death
- Severe digestive symptoms

Eight foods account for 90 percent of allergic food reactions. These are milk, eggs, fish, shellfish, peanuts, tree nuts, wheat and soybeans. People who have food allergies must strictly avoid these foods.



# FOOD SENSITIVITY/INTOLERANCE

Food sensitivities and intolerances are more common than food allergies. A food triggers an intolerance in your **digestive tract** when your body can't properly break it down, or your body reacts to a food you're sensitive to. Food sensitivities or intolerances do not involve the immune system.

Intolerance reactions are generally less severe than allergic reactions and are unlikely to be life threatening. However, intolerance symptoms may still cause discomfort.

**You may be sensitive or intolerant to a food for a few reasons. These include:**

- Not having the right enzymes you need to digest a certain food. For example, lactose intolerance is when your body can't break down lactose, a sugar found in dairy products, because you lack an enzyme called lactase
- Consuming food additives (pesticides, colourants, preservatives, flavourants and flavour enhancers such as MSG)
- Malabsorption of sugars naturally found in certain foods such as fructans, galactans, fructose, malitol and sorbitol

## Symptoms of food sensitivity:

- Ongoing low energy levels and chronic fatigue

### • Respiratory issues:

- Asthma, sinusitis
- Post nasal drip, runny nose

### • Digestive issues:

- Nutrient deficiencies, including anaemia (iron deficiency)
- Gas and bloating, diarrhoea, constipation, reflux, heartburn, vomiting, pain, cramping and nausea.

### • Nervous system issues:

- "Brain fog," difficulty concentrating and trouble remembering information, hyper-active behaviour
- Frequent headaches or migraines
- Mood-related changes, including anxiety, panic attacks and increased depression symptoms

The most common food intolerances are for **gluten** (a protein found in grains such as wheat, barley, oats and rye) and **lactose** (a milk sugar). Intolerances are also found for chocolate, strawberries, citrus fruit, tomatoes, egg, corn, yeast, and soy.

Elimination diets remove foods most commonly associated with intolerances for a period of time until symptoms subside. Foods are then reintroduced one at a time while monitoring for symptoms.



# GLUTEN FREE

## What is Gluten?

Gluten is a protein found in certain grains such as wheat, barley, spelt and rye. It's the sticky part of the grain responsible for the elasticity of the dough, and is commonly used as a bulking agent in many modern food products. Wheat has been selectively bred over hundreds of years to contain an increased amount of gluten.

## Gluten intolerance

It is becoming very common for people to experience symptoms after eating gluten. This could be a sign that the person is struggling to digest the gluten protein found in these grains. We term this 'Gluten Intolerant' and suggest these people choose naturally gluten free or low gluten products.

## Coeliac disease

On a more serious side of the scale are those who suffer from Coeliac disease. These people cannot eat any gluten at all. Coeliac Disease is an auto-immune disease where, upon exposure to gluten, the body exhibits an abnormal immune response. In the small-bowel this causes an inflammatory reaction and may produce shortening of the villi lining the small intestine. This interferes with the body's ability to digest and absorb food and can cause nutritional deficiencies, general fatigue, joint issues and gastrointestinal symptoms. The only treatment is through diet, cutting out all gluten containing foods.

## What about oats?

The story with oats is that they are naturally gluten free, but are often contaminated with gluten containing grains. For this reason, it is advisable that people with coeliac disease avoid normal oats and rather opt for the certified gluten free oats. If oats have a dedicated growing, processing, storage and transport chain then they can be labelled as gluten free, which is why we have some oats that are and some that are not. People with Gluten Intolerance might be able to tolerate normal oats, as their gluten content is very low.

## Eating gluten free at Wellness

Wellness sells a range of gluten free breads, breakfast cereals, pizza bases and pastas, but you can also make these items yourself using fresh ingredients and naturally gluten free grains, beans and legumes. Paleo and Banting Products are also gluten free.

## Gluten free grains:

- |                    |                    |
|--------------------|--------------------|
| - Rice             | - Teff             |
| - GMO free corn    | - Amaranth         |
| - GMO free polenta | - Millet           |
| - Buckwheat        | - Sorghum          |
| - Quinoa           | - Gluten free oats |

## Baking substitutes:

There is no need to stop baking when on a gluten free diet.

Wellness offers many alternatives that work perfectly in baking:

- Gluten free flour mix
- Chickpea flour
- Tapioca flour
- Cassava Flour
- Almond flour
- Coconut flour
- Buckwheat flour
- Brown rice flour



**Important:** Be mindful that gluten free, ready-made and convenience products may still be very refined and may contain added sugars and artificial additives such as binders, preservatives and flavour enhancers.

## LACTOSE FREE

Dairy free products will contain no ingredients made from the milk of any animal, whether cow, goat or sheep. This includes milk, butter, yogurt, cheese and kefir. Eggs are not included in dairy products.

### GLOSSARY OF DAIRY

**Casein:** The main protein in milk. The type of casein found in cow's milk is not easy to digest in its natural state, but becomes more digestible when made into yoghurt or cheese.

**Curds:** This is the milk protein part of milk that separates out in cheese making, containing mostly casein and fat.

**Kefir:** A thin fermented milk drink. Fermented using a starter culture of bacteria and yeasts, it is like a thin, slightly sour yoghurt.

**Lactase:** The enzyme that digests lactose. Lack of lactase can cause a lactose intolerance.

**Lactose:** The main sugar in milk. In those who cannot digest it, it causes bloating and digestive discomfort.

**Pasteurisation:** A method by which milk is sterilised to get rid of the naturally occurring bacteria. This increases its shelf life and is done to make it safer.

**Skimmed milk:** Skimming refers to taking the cream off the top of the milk to lower the fat content.

**Yoghurt:** Milk that has been fermented by adding bacterial cultures. These bacteria produce lactic acid when breaking down the lactose.

**Whey:** The liquid that remains after separating milk into curds during cheese-making. It contains the left-over protein and is often used as a protein supplement.

### LIVING LACTOSE & DAIRY FREE

#### Lactose intolerance

Lactose is the main sugar found in all milk and it is digested in our intestines by an enzyme called lactase. Lactase breaks down lactose so that it can be absorbed into the blood stream. The levels of the enzyme drop as we get older and some people are predisposed to lower levels. If people do not have enough of this enzyme, then eating lactose-containing foods can cause bloating, digestive discomfort and diarrhoea.

#### Cows, goats and sheep

Cow's milk in particular can be difficult to digest for lactose intolerant people. A lot of people who find cow's milk difficult to digest have no problem with goat's milk. Goat milk contains less lactose than cow's milk which makes it easier to digest. However, if someone is lactose intolerant then they should also avoid goat milk and sheep milk.

#### Cheese and yoghurt

Yoghurt is fermented, which means bacteria have started breaking down the sugar in the milk. This reduces the lactose content. Likewise when making butter the fat is separated from the sugar, so butter is very low in lactose, while in cheese-making the lactose is drained away in the whey, and so does not end up in the cheese. For anyone with a complete lactose intolerance, absolutely no dairy products are suitable unless they have specifically had the lactose removed, but for those with a milder intolerance it is possible that it is just milk that is the problem. In this case, yoghurt, butter, ghee or hard cheese in small quantities may be well tolerated.

There are some people who have an immune reaction to the casein protein in cow's milk. An allergy is very different to an intolerance, and rather than causing digestive problems it can cause rashes and difficulty breathing. As with all serious allergies, those with a milk allergy need to avoid all foods that contain even minute traces of dairy products.

- Coconut milk
- Almond milk
- Oat milk
- Soya milk
- Coconut yoghurt
- Cashew nut yoghurt
- Frozen dairy free deserts
- Dairy free cheese



1. What is the difference between a food intolerance and food allergy?
2. What are the two most common food intolerances?
3. What is gluten?
4. Name 8 gluten free grains that Wellness Warehouse sells?
5. What flours may be used for gluten free baking?
6. What is lactose?
7. What is casein?
8. List 5 dairy free alternatives you can offer a Wellness Warehouse customer?

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## 3. UNDERSTANDING LABELS

### ORGANIC FOOD

The move to organic food is becoming increasingly popular worldwide due to the enhanced nutritional benefit that can be obtained from eating organic foods. South Africa is somewhat behind on this trend due to the increased costs involved in organic farming, and organic food. However, we are following the rest of the world and Wellness Warehouse strives to be a leader in the Organic food market in South Africa.

#### What is organic farming?

Organic production systems are designed to produce optimum quantities of foods of high nutritional value by using practices which aim to avoid the use of chemicals and which minimise damage to the environment and wildlife.

#### Let's look at this definition a bit more closely:

Produce optimum quantities of food - aims to grow a sustainable amount of food from an area of land, and not to deplete the soil quality by over-farming.

High nutritional quality - by providing the soil with high levels of absorbable nutrients in the form of natural fertilisers and using methods such as crop rotation, the soil is protected against depletion of important minerals and trace minerals so that organic crops contain higher levels of nutrients.

Avoid the use of chemicals - natural fertilisers are used to maintain soil fertility and natural methods are used to control pests and weeds.

Minimise damage to the environment – by avoiding the use of toxic pesticides and herbicides organic farming is better for the local environment and wild-life.

In order to sell a food as organic, every step of the chain must be certified, including the transport, storage and production. Organic products will be marked with a certification symbol and will be easily identifiable on the label.

### CERTIFIED ORGANIC

A product that has gone through extensive testing and auditing processes and has been certified organic by a professional body. The item must have an ingredients list and the contents should be 95% or more certified organic, meaning free of synthetic additives like pesticides, chemical fertilizers, and dyes, and must not be processed using industrial solvents, irradiation, or genetic engineering.

The remaining 5% may only be foods or processed with additives on an approved list. Livestock are reared in as natural an environment as possible without routine drug injections. The certification body will always be shown on the product label.



## FAIRTRADE

The Fairtrade Label is the leading sustainability stamp in the world. Fairtrade ensures a better deal for small-scale farmers, farm workers and the environment in developing countries.

Products bearing the Fairtrade Label are audited against Fairtrade Standards which cover:

- Labour Rights,
- Environmental Standards, and
- Economic Standards - the Fairtrade Development Premium and the Fairtrade Minimum Price.

The focus of Fairtrade in South Africa is to create a more equal and sustainable Africa. The label empowers consumers and businesses by giving them a choice to purchase ethical products.

Look for the Fairtrade symbol on your favourite Wellness products.



## KOSHER/HALAAL

Some customers because of religious beliefs are only able to eat food that has been deemed fit to eat by their religious body. It is important to respect the customers right to choose and to show them which foods and supplements would be appropriate. Food items will display the relevant labels.

- Jewish customers may want food/supplements that are Kosher/Parev.
- Muslim customers may only want food/supplements that are Halaal.



## QUESTIONS:

1. What does it mean to farm organically?
2. What does certified organic mean?
3. Name 4 food products that Wellness Warehouse sells that display certified organic symbols?
4. Why is Fairtrade important?
5. Which customers would be looking for Kosher or Halaal products?

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## 4. SUPERFOODS

### What is so super about Superfoods?

A superfood is basically any nutrient dense or antioxidant rich food. Including superfoods in meals or recipes is a good way to increase our daily nutrient intake. Superfoods can be eaten whole, mixed into smoothies, sprinkled on salads or added into cooked foods that, when added to the diet, has immense health benefits.

### What are antioxidants?

Antioxidants are plant chemicals found in foods that have a wide variety of health benefits. They prevent cell damage, therefore protecting cells and helping with regenerating of cells. Due to their protective nature, many studies have shown their anti-ageing and anti-cancer properties. Foods rich in antioxidants are often referred to as superfoods, as they are SUPER for your health.

## COMMONLY USED SUPERFOODS

### Aloe

Aloe is usually used either topically as a gel or as a drink. It can help to detoxify the liver and colon. It helps to clear up acne and eczema, regenerates the skin and improves elasticity of skin and joints. It also can be used to reduce inflammation and promote healing for the gut.

### Baobab

Is an African superfood. Because it is rich in minerals it helps to alkalise the body. It is also a good source of Vitamin C which benefits the immune system. High in soluble fibre which can help keep good gut flora healthy.

### Cacao nibs or powder

The raw ingredient from which chocolate is made, cacao beans are thought to be one of the most nutrient rich foods on earth. High in antioxidants, magnesium and mood boosting chemicals.

### Chia seeds

It is said that Aztec warriors would consume a mixture of Chia and water during hunting trips to give them stamina and endurance. A gluten free super seed, Chia Seeds contain protein, vitamins and minerals and essential vegetarian omega-3 fatty acids, as well as fibre.

### Goji Berries

In Chinese Medicine, Goji Berries are traditionally used as a longevity and vitality enhancing fruit. This tiny red antioxidant superfruit is packed with vitamins and minerals and are a delicious addition to smoothies and trail mixes.

### Grasses

Wheatgrass and barleygrass are the better known grasses, traditionally they are juiced from fresh leaves. They are naturally gluten free. Nowadays this juice is dried and can be taken in powder form. Grasses are nutrient rich and contain essential enzymes that help to stimulate the digestive system. They are also cleansing and alkalising as well as helping to oxygenate the blood. They are essential for healing the stomach and entire digestive tract.

### Hemp seeds or powder

Shelling the hemp seed removes the tough outer shell. This process does not compromise the nutrition obtained from eating these super seeds. Hemp Seeds are the perfect protein for vegans and vegetarians as they contain all the essential amino acids and provide essential omega-3 fatty acids.

**Lucuma**

Powdered Lucuma is an exotic South American superfruit with a host of nutrients, minerals and antioxidants. Lucuma Powder has a sweet maple syrup taste which means it can be enjoyed as a natural sweetener.

**Maca**

Maca powder, also known as Peruvian ginseng, is an adaptogen - it "adapts" to your body's metabolism, working with your body for optimum results. As an adaptogen, Maca can provide more energy if it is needed, but if it is not, it will not overstimulate. It is often used to balance hormones. Black is more masculine, white more feminine.

**Matcha**

Is finely ground high-quality green tea, it is traditionally used for Japanese tea ceremonies. It is a potent anti-oxidant containing L-Theanine which helps keep the brain calm and focused. Also helps improve energy, speeds up recovery post exercise and aids weight loss.

**Mesquite**

Mesquite powder is made from the pods of the mesquite plant. This superfood is packed with vitamins, minerals and antioxidants. Mesquite Powder has a sweet, delicate flavour and is the perfect addition to smoothies and desserts.

**Moringa powder**

A superfood from Northern India. It contains high amounts of vitamin A and C, as well as the minerals potassium, iron and calcium. It can be used during pregnancy to provide much needed extra nutrition. It may help to heal the digestive system and can also be used to help recovery post exercise. A natural multi-vitamin and mineral supplement.

**Spirulina powder or tablets**

Spirulina is a blue-green algae traditionally used for its effective cleansing and detoxifying effects. Spirulina is rich in vitamins, minerals, essential omegas and a source of high quality, easily digestible vegetarian protein.

**White Mulberries**

White Mulberries are traditionally used in Asian medicine as a blood tonic. This small superfruit is packed with minerals and fibre. White Mulberries have a mild, sweet taste and are a delicious addition to smoothies and trail mixes.

## QUESTIONS:

1. What are some of the benefits of aloe taken as a drink?
2. What important mineral is found in raw cacao?
3. How may you suggest a customer use chia seeds?
4. Why would someone want to include hemp seeds in their diet?
5. Name 3 superfoods that are good for energy and stamina?
6. Which superfoods could a customer use instead of a multivitamin?
7. Which two superfoods are very high in Vitamin C?
8. Which 3 superfoods are naturally sweet and good to add to smoothies?
9. Which superfoods are very healing to the stomach and digestive tract?

## 5. DIETARY TRENDS

### VEGETARIAN & VEGAN

**Vegetarian diet** - Contains no meat, poultry, game, fish. May contain eggs, dairy products and honey.

**Vegan diet** - Contains no ingredients of animal origin, including meat and fish as well as eggs, dairy products and honey.

#### Why people make the choice to be vegetarian or vegan:

- **Ethical reasons**

To keep up with consumer demands for animal products, farming practices have been compromised. Animal cruelty in intensive farming systems may influence a person to put their foot down and 'boycott' this industry. Vegetarians may be swayed into eating products from animal origin (eggs, dairy products) if it has not harmed the animal in any way.

- **Environmental reasons**

Plant-based foods tend to use less water, require less land for production, and emit fewer greenhouse gases than the production of commercial animal products.

- **Health reasons**

People have reported improved health after following a plant-based diet. This could be due to the amount of added growth hormones and antibiotics found in animal products that come from intensive animal farming. Furthermore, animal products take longer to digest and puts more strain on our digestive system.

#### VEGETARIAN AND VEGAN LIVING:

Foods on the shelf are often marked as 'suitable for vegetarians' or 'suitable for vegans'.

As with all diets, it is possible to have either a very good or a very bad diet as a vegetarian or vegan. To obtain the full spectrum of amino acids and nutrients, it is important to eat a **varied diet**, including a mixture of whole grains, beans and legumes as well as fresh fruit and vegetables.

It is important to include some **good quality protein sources** when following either of these diets. Vegetarians can include dairy products and eggs, while vegans can get all the protein they need from protein-rich beans, pulses (pea protein), nuts and seeds (hemp protein), superfoods such as spirulina and chia seeds, and to a lesser extent from grains.

It is common for vegetarians and vegans to be deficient in **vitamin B12** and supplementation is recommended. There are no proven good plant sources of vitamin B12 and while vegetarians can rely on dairy products, vegans are best advised to take it as a supplement. A deficiency of vitamin B12 can cause anaemia, or a shortage of red blood cells, the initial symptoms being fatigue, tingling in the hands and feet and looking very pale.

There are some **hidden sources of animal products** such as gelatine, used as a thickener in jellies and desserts, and meat stocks used in ready-made soups.

Recommend vegan or vegetarian supplements with capsules made from plant cellulose (from pine or plantar trees) and not gelatin (from bovine or porcine sources). Make sure you know your Wellness products so that you can offer the best vegan or vegetarian **Omega 3** (from plants and not fish oils) and **vitamin D** (from lichens and not lanolin) options to customers.

Wellness also offers a range of dairy free (vegan-friendly) products as well as vegan 'meat' products.

## RAW FOODS

A food is considered raw if it has never been heated over 46°C. A raw food diet is usually plant-based, being made up mostly of fruits, vegetables, nuts, seeds, sprouts and plant oils. Raw foods should not be refined, pasteurised, treated with pesticides or otherwise processed in any way.

The idea behind eating raw is that the more we cook and prepare foods, the less nutritious they become. Vitamins, antioxidants and enzymes are very temperature sensitive and start to denature or break down above about 46°C. Raw foods help alkalise the body and reduce acidity.

People that eat raw food claim to have higher energy levels, clearer skin and a greater sense of vitality on a raw food diet. Some foods, such as oats and nuts, are better digested when they have been either soaked or cooked. Raw foodists generally soak and sprout a lot of their food, and often use dehydrators to 'cook' their food.

The easiest way to experience the health benefits of raw foods is to eat more salad, fruit and smoothies, snack on raw nuts and try sprouted grains and seeds.

### **Suitable for:**

- Persons who have chronic or serious illnesses as a reset
- Persons who would like to detoxify their bodies

### **Might not be suitable for:**

- For some it can be harder to digest raw food and they may need a digestive enzyme
- May not be suitable for everybody or long term, especially in winter, as it cools down the metabolism

## LOW CARB HIGH FAT (LCHF) DIETS

This way of eating helps your body shift from burning carbohydrates (glucose) for energy to burning fat. The ideal ratio between the macronutrients to achieve this is to eat a low carb, medium protein and high fat diet.

### **Banting**

Banters eat mainly meats and fats such as olive oil, avocado and full cream variants of dairy. It minimises carbohydrate-based foods such as grains, fruits, sugar and root vegetables.

### **Ketogenic**

A more extreme version of Banting where carbohydrates are restricted to less than 50g per day so that the body starts to use ketones for energy. The ketones are produced in the liver from eating a diet high in healthy fats.

Persons who are obese, overweight, who suffer from epilepsy, cancer, diabetes or insulin resistance might benefit from these diets. Recommend that these diets be followed under a health care provider's supervision.

### **For these diets to be successful make the following recommendations to customers:**

- Focus should be on medium, and not high, protein
- Minimise animal fats, include more healthy plant fats
- Make sure proteins and fats come from clean sources (wild caught, grass-fed, free range and organic)
- Include 7 to 10 cups of low starch leafy vegetables for added nutrients and fibre
- Combine with Intermittent Fasting to obtain best results



# QUESTIONS:

1. What foods do vegans avoid?
2. What alternatives can you offer a vegan customer who wants to replace milk, cheese or yoghurt in their diet?
3. Name three plant-based proteins a vegan customer may want to consider adding to their diet?
4. Why would eating a raw diet be beneficial?
5. What healthy foods would a customer following a raw food diet eat?
6. What happens to energy production in your body when you eat a low carb and high fat diet?
7. What is the ratio of macronutrients in a LCHF diet?
8. What is the difference between the Banting and Ketogenic diets?
9. Persons with which illnesses would benefit from a LCHF diet?
10. What five recommendations would you make to a customer wanting to follow the Ketogenic diet successfully and healthily?

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

# 2

## MODULE TWO

### BODY CARE

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*"Your skin is a really great indicator of what is going on inside your body."*

- Sami Blackford

# 1. UNDERSTANDING NATURAL SKIN CARE

## OUR SKIN HAS FIVE IMPORTANT FUNCTIONS:

- **Protection** - The skin is our first line of defence against mechanical impact, microbes and radiation. The top layer of skin is covered with a thin protective layer known as the acid mantle. This is formed by free fatty acids secreted by sebaceous glands and lactic and amino acids from sweat. It is very important for the normal healthy functioning of the skin to preserve this protective mantle which keeps moisture in and organisms (such as bacteria, viruses and fungi) and toxins from entering the skin. This sebum layer also protects the skin from drying out. The skin also has a natural microbiome and it is important to maintain this to ensure healthy skin. Beneficial bacteria on the skin help improve skin hydration, reduce sensitivity and slow down ageing. Harsh personal care products can damage these bacteria and may contribute to many chronic skin conditions. Melanin in the skin protects cell DNA from UV light. The epidermis also has Langerhans cells, which help to regulate immune responses to pathogens that come into contact with the skin.
- **Thermal regulation** - Temperature regulation is aided by the skin through the sweat glands and blood vessels in the dermis. Increased evaporation of secreted sweat decreases body temperature. Vasodilation (relaxing of small blood vessels) in the dermis makes it easier for the body to release some heat and lower the body temperature through skin. In vasoconstriction (contracting small blood vessels), the dermis retains some of the internal body temperature. The fatty subcutaneous layer of the skin also acts as an insulation barrier, helping to prevent the loss of heat from the body and decreasing the effect of cold temperatures.
- **Sensation** - An important function of the skin is to detect the different sensations of heat, cold, pressure, contact and pain through the nerve endings in the dermis.
- **Excretion and absorption** - Sweat carries salts, ammonia and urea for excretion and is acidic which has an antimicrobial function. At the same time, our skin, while acting as a barrier to the outside world, does absorb much of what we put onto it. This means that we may be absorbing up to 100 chemicals just from our body care products. The expression "If you cannot eat it don't put it on your skin" is an effective way to decide which products to use.
- **Endocrine function** - Skin is one of our main sources of vitamin D, through the production of Cholecalciferol (D3).

## WHY NATURAL SKIN CARE?

- Natural skin care products are gentle on our skin and are particularly beneficial for anyone with sensitive-, dry-, dehydrated-, problem- or ageing skin
- They do not contain any harmful chemicals and are made from natural plant extracts (plant oils, essential oils and botanical extracts)
- They do not strip the skin from natural oils and beneficial bacteria and they protect the integrity of the acid mantle on the skin
- They are gentle on the environment as they can be biodegradable
- They are cruelty free as they are not tested on animals

### Body care products extend beyond the skin and also include:

- Hair products - shampoo, conditioners and hair treatments
- Toothpastes - fluoride free options and toothpastes with plant extracts
- Personal feminine hygiene - tampons and sanitary pads made from organic cotton and which are chemical free



## READING BODY CARE LABELS:

### Natural

For the natural cosmetic label the product must contain a minimum of 50% of plant-based ingredients and a minimum of 5% of all ingredients by weight must come from organic farming.

### Organic

For the organic label the product must contain a minimum of 95% plant-based ingredients and a minimum of 10% of all ingredients by weight must come from organic farming. The product has been certified organic by a professional body such as Ecocert or SOIL.

### Vegan

A product that is vegan is not only cruelty-free but does not contain ingredients derived from any animal sources or ingredients from bees (honey, beeswax, propolis). The term cruelty-free typically means there was never any testing done on animals at any stage of a product's creation, manufacture or distribution.

### Anti-oxidants

Antioxidants protect the skin by limiting the production of free radicals, which can damage skin cells. Antioxidants in skin care products can do a lot for the health and appearance of your skin, including reducing the signs of aging. Many plant extracts and oils are a rich source of natural antioxidants.

### Gluten free

On skin care ingredient labels, primary sources of gluten are listed as hydrolyzed wheat protein, triticum vulgare (wheat germ) oil or extract. A gluten free body care product has been tested by a laboratory and certified to be gluten free.

### Cruelty free

Many commercial skin care products are tested on animals to check for allergic reactions. These practices can cause great discomfort and pain to the animals involved. PETA certified (bunny) means skin care products have not been tested on animals.



Recyclable



Biodegradable



Compostable

## INGREDIENTS TO AVOID

Natural skin care products should not contain the following chemicals:

### Parabens

Parabens are common cosmetic ingredients used to preserve the products. Studies have shown that these parabens may cause hormones to be disrupted and potentially contribute to adverse health effects. Paraben-free products do not contain these potentially toxic ingredients.

### SLS (Sodium Lauryl Sulfate)

A widely used and inexpensive chemical found in many mainstream personal hygiene products such as shampoos, toothpastes, mouthwashes, soaps and body washes, along with Sodium Laureth Sulfate (SLES) and Ammonium Laurel Sulfate (ALS). It strips the skin disrupting the pH, making the skin more dry and sensitive. It helps other chemicals get into the body. Many customers are looking for SLS-free body care products or may have damaged skin due to using SLS for many years.

## Petrochemicals

Almost all personal care products contain by-products of the petrochemical industry; petroleum jelly, mineral oil, petrolatum and paraffin oil. The skin cannot absorb these chemicals and they remain on the surface giving the feeling of moisturising but can upset the skin's natural oil balance.

## Phthalates

These are used to keep the fragrance/parfum chemicals in solution and are potential hormone disrupters. They enter the blood stream and are stored in fat cells where they build up in the body.

## Aluminium

Is used in anti-perspirant deodorants and blocks the sweat glands in the armpits. This can block the natural detox process. Aluminium is also a neurotoxin where it crosses the blood-brain barrier and can be a factor in Alzheimers disease.

## Formaldehyde

Is used to preserve bodies and organs. It is a known carcinogen and is often found in makeup. It can cause an allergic reaction.

## Peroxide

Is used to soften the hair cuticle and penetrates the hair shaft where the hairs natural pigment (colour) can be lightened. Hence the name peroxide blonde. Peroxide is a strong chemical that oxidises the hair causing the hair to become dry and weak.

## Ammonia

This chemical is used along with peroxide to help the colour get into the hair shaft. It causes the hair to lose moisture. It damages the pH of the hair and scalp. Some customers are sensitive to ammonia and may have a severe reaction to it.

## MORE CHEMICALS TO AVOID

- PEG
- Talc
- DEA, TEA & MEA
- EDTA
- Urea
- Lead

## NATURAL BEAUTY BRANDS

Customers looking for natural skin care brands can be recommended the following options:

- Dr Hauschka
- Esse
- Skoon
- REN Clear Skin
- Pradance



Other Natural skin care ranges to suggest are:

African Organic, Alba Botanicals, Back2Nature, Badger, Beaucience, Dr Organic, Earth Sap, Eli & You, Jasons, Natural Beauty, Naturalmente, Oh Lief, Pure Beginnings, Simply Bee, The Natural Family Co, Victorian Garden



## 2. THE BENEFITS OF NATURAL BODY CARE PRODUCTS

### SENSITIVE SKIN

- Customers with sensitive skins may react to harsh chemicals and even potent essential oils in some products.
- Natural plant oils and botanicals can help alleviate sensitivity.
- Probiotics in natural products may assist in reestablishing a healthy microbiome which may alleviate sensitivity.

### DRY/DEHYDRATED SKIN

- Dry and dehydrated skin is very common in South African due to our harsh climate.
- Natural plant oils such as argan-, almond-, avocado- and marula oils and butters such as shea butter, cocoa butter, and jojoba butter may assist in nourishing dry skin.
- Products containing aloe vera and hyaluronic acid may assist to rehydrate dehydrated skin.

### PROBLEM SKIN

- Often customers with acne or pimples use harsh cleansers which strip the natural oils forcing the skin to produce more oil.
- Natural skin care products are gentle and cleanse without stripping the skin of natural oils and beneficial bacteria.
- Natural plant extracts such as tea tree, lemon and lavender essential oils inhibit infections, while rosehip oil and helichrysum-, frankincense-, geranium- and lavender essential oils will assist in healing scars.

### AGEING SKIN

- Our skin becomes damaged from oxidative stress caused by excessive sun exposure, smoking and poor diet. Our skin also loses collagen as we age.
- Signs of an ageing skin are: sun/liver spots, wrinkles, loose and sagging skin.
- Natural plant oils are rich in anti-oxidants helping to slow down the ageing process.
- Remember to recommend anti-aging supplements that contain collagen, fatty acids and anti-oxidants.

### SHAMPOO

- Commercial shampoos contain sodium laurel sulphate which strips the hair of natural oils making it feel squeaky clean but leading to dry and damaged hair.
- Natural shampoos are gentle on the scalp and hair and may help to improve the health of hair and scalp.
- Can be beneficial for those suffering from dandruff and scalp eczema.
- Hair may go through an adjustment period as it detoxes and adapts to natural ingredients. Natural shampoos might not foam as much as conventional shampoos.

### CONDITIONERS

- Natural conditioners nourish the hair and scalp.
- They can be used as a leave in styling product.

### HAIR DYES

- These are more gentle on the hair and scalp and don't damage the hair and scalp like regular hair dyes do.
- Some products may still use hydrogen peroxide but in far less amounts.
- Other hair dyes use only completely natural ingredients with no ammonia or peroxide.

## NATURAL TOOTHPASTES

- These toothpastes generally don't contain flouride and foam less, commercial toothpastes can damage the mouth's natural flora.
- Natural toothpastes use plant extracts to help keep the mouth and gums healthy.

## FEMININE HYGIENE PRODUCTS

- Commercially available tampons and sanitary pads are made from bleached cotton that has gone through chemical processes. They also contain plastic and fragrances that may be irritating to delicate skin. Cotton that is not grown organically may contain pesticides or may be genetically modified.
- Women that are prone to vaginal thrush or general vaginal irritations and sensitivity would benefit from using organic personal hygiene products.
- Also consider re-usable products such as the moon cup and fabric pads.

## QUESTIONS:

1. Which requirements should be adhered to, to label a product as natural?
2. When would a product be certified organic?
3. What does it mean when a product is vegan?
4. What symbol will show when a product is cruelty free?
5. Name four ingredients that are best to avoid in personal care products?
6. How may a customer with sensitive skin benefit from natural skin products?
7. Why are natural products better for problem skins?
8. What does ageing skin need to help slow down the ageing process?
9. What may you want to warn a customer about when they move to a natural shampoo?
10. Why are natural toothpastes better?
11. Why would you suggest a customer try natural sanitary ware products?
12. Which re-usable feminine hygiene products can you recommend?

This image shows a single sheet of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

# 3

## MODULE THREE

### ECO-HOME

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*"Being green is more than just buying 'eco'. It is an unshakable commitment to a sustainable lifestyle."*

- @ecowarriorprincess

# 1. UNDERSTANDING NATURAL HOME CARE

We don't always realise how toxic our homes can be, as the cleaning products we use often contain harmful chemicals. These chemicals are not only absorbed through the skin but they can also be inhaled. They can lead to an increase in your body's toxic load and as many of these chemicals are stored in fat cells they accumulate in our body over time.

Using natural and chemical free cleaning products can help to reduce our toxic load.

## WHY NATURAL HOME CARE?

### Natural home care products:

- Contain natural ingredients - no harmful chemicals
- Are Bio-degradable - they break down in the environment
- Cruelty free - have not been tested on animals
- Fragrance Free - only natural fragrances
- Hypo-allergenic - for sensitive skins



## SAFE & ECO FRIENDLY HOME CARE PRODUCTS

- Cleaning products
- Laundry products
- Air fresheners
- Natural incense
- Air purifiers
- Water purifiers
- Plastic free or BPA Free options - straw, cups, drinking bottles, baby bottles
- Kitchen appliances - juicers, blenders and dehydrators
- Healthy Pet Care

## RECOMMEND NATURAL HOME PRODUCTS FOR:

- Customers with chemical sensitivities who react to chemicals in cleaning products and are looking for products that are hypo-allergenic and contain natural ingredients.
- Customers suffering from skin complaints such as dry skin, eczema, skin rash or hives and can't use regular washing machine powders and softeners which can irritate the skin.
- Customers wanting cleaning products that are gentle on the skin, hands and don't damage nails.
- Customers who are concerned about the environment and are looking for alternatives to plastic and BPA free products and prefers biodegrade, compostable and eco-friendly products.
- Customers who are looking for products that have not been tested on animals and are vegan.
- Customers who want to avoid hormone disrupting chemicals, for example those who suffer from pms, infertility and during menopause.
- Customers who want to lessen their toxic load, for example those who have cancer.



## INGREDIENTS TO AVOID

- **BPA** - is the name for bisphenol A, a chemical used when making plastic, it mimics the hormone oestrogen. Food and drink stored or heated in BPA plastic can be affected as the BPA leaches out.
- **Phthalates** - found in plastic, mimics the hormone oestrogen and has been linked to some cancers, may affect male fertility.
- **Ammonia** - gives that sparkle to counters and stoves but is a powerful irritant and can cause lung and breathing issues eg chronic bronchitis and asthma.
- **Bleach/Chlorine** - the fumes can be inhaled and is a respiratory irritant
- **PERC (Perchloroethylene)** - is found in carpet and upholstery cleaners, it is a neurotoxin and possibly carcinogenic
- **Triclosan** - is a powerful anti-bacterial found in kitchen and toilet cleaners, as well as hand sanitisers. It is a hormone disruptor and carcinogenic, it may promote drug resistant bacteria.
- **QUARTS (Quarternary Ammonium Compounds)** - found in fabric softeners, it is anti-bacterial and can be a skin irritant.
- **2-Butoxyethanol** - an alcohol found in window cleaning products, when it is inhaled it can cause a sore throat.
- **Sodium hydroxide** - found in drain and oven cleaners, it can burn the skin and eyes on contact and cause a sore throat if inhaled.
- **Formaldehyde** - used as a preservative in household products, research shows it may be carcinogenic. It may also be linked to joint pain, headaches and chronic fatigue.
- **Artificial dyes, colours and fragrances**

## GREEN CLEANING BRANDS

- |                                |                 |
|--------------------------------|-----------------|
| • Wellness Warehouse Own Brand | • Better Earth  |
| • Ecover                       | • Earth Sap     |
| • Method                       | • Triple Orange |



## QUESTIONS:

1. Give 5 reasons why a customer may want to try natural cleaning products?
2. Explain why customers should avoid BPA in plastic products.
3. Discuss why Triclosan is detrimental to your health.

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# 4

## MODULE FOUR LIFESTYLE

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*"In many shamanic societies, if you came to a shaman or medicine person complaining of being disheartened, dispirited, or depressed, they would ask you one of four questions: When did you stop dancing? When did you stop singing? When did you stop being enchanted by stories? When did you stop finding comfort in the sweet territory of silence?"*

- Gabrielle Roth

Lifestyle factors play an important role in the health of the Nervous System.

When we stress, we engage our **Sympathetic Nervous System** – this is our flight, fight or freeze response. If the body stays in this stressed state too often or too long it can lead to poor digestion, poor sleep and a compromised immune system.

Lifestyle factors are important for engaging the **Parasympathetic Nervous System** - this system helps us to relax, to rest and digest and bring the body back into balance.

## EXERCISE

Exercise should be regular, enjoyable and varied. Ideally you should aim to strike a balance between aerobic exercise, core and strength training, while maintaining flexibility and mobility. Weight and resistance training assists in building strong bones.

Daily, gentle exercise is important (walking, swimming, rebounding, gardening) to encourage bowel movements and lymph drainage, as well as cope with stress and low mood.

High-intensity interval training (HIIT) is popular as it provides excellent results and requires very little time in a busy day. It involves short, intense bursts of exercise with less intense moves or complete rest in between.

Restorative yoga helps to relieve mental and physical fatigue and soothes the sympathetic nervous system.

Very strenuous exercise such as training for marathons will produce a high stress response in the body and care should be taken to support the body with extra nutrients, adaptogens and rest.

## SLEEP & REST

It is of utmost importance to get sufficient and restful sleep. During the night, the body repairs, restores, heals, maintains and detoxifies itself. Sleeping well will boost immunity and weight loss.

Establish a bed-time routine that ensures you sleep at least 7 or 8 hours a night. Avoid blue light from electronic devices two hours before going to bed and keep the room as dark as possible. Avoid caffeine, alcohol, tobacco and any other stimulants before bed-time.

Getting the body into a good sleep routine begins in the morning. Studies have shown that getting early morning sunlight can help address insomnia by helping to establish a healthy circadian rhythm. Use breathing techniques, visualisation and meditation to calm the nervous system before bed.

Make time every day to relax and unwind – take a walk with the dog, create a lovely dinner or sit outside with bare feet on the grass. It is important to digest the emotions experienced during the day before going to bed.



## NATURE

Being in nature, or even viewing scenes of nature, reduces anger, fear, anxiety and stress and increases feelings of self-confidence and joy. Exposure to nature not only makes you feel better emotionally and mentally, it contributes to your physical wellbeing, reducing blood pressure, heart rate, muscle tension, and the production of stress hormones.

Spend time outdoors every day to connect with nature and get some much needed vitamin D at the same time. Walk with bare feet on the earth to ground your body and unload inflammatory electrons accumulated through the day.

## DETOX

The toxins in our environment accumulate over time and burden our elimination organs, especially our kidneys and liver. Take steps to detoxify your environment to give your body the best circumstances to function optimally.

Most personal care products and make-up, as well as household cleaning products, contain many harmful chemicals and toxins. They can be absorbed through our skin and may increase our toxic load and are detrimental to our health. Switch to natural and non-toxic products.

Make sure fruit and vegetables are organically grown to avoid pesticides, and ensure meat, eggs and dairy products are obtained from pasture raised and free range animals to avoid growth hormones and antibiotics. Drink only filtered water and avoid bottled and tap water to minimise chlorine, xeno-oestrogens and heavy metals.

The radiation from electronic devices may also add up to our toxic load – be mindful to lessen time spent on cellphones, laptops and tablets, and minimize time spent in wi-fi areas. Be mindful of what you consume in terms of social media, news, television, music and movies. Make sure the conversations, people and energy you allow in your life are healthy and supports your highest good.

## CONNECT

We are a disconnected society which is reflected in how we treat the earth, ourselves and others – it is the reason we have so many ills and illnesses in the world. Make time to:

- **Connect with Yourself:** Do something creative and self-nurturing regularly
- **Connect with Others:** Be of service to someone in need, reach out to strangers with acts of random kindness
- **Connect to your Community:** Find your tribe at the gym, school, church or work
- **Connect to Nature:** Take up gardening, grow your own veg, hike or spend time in nature

## STRESS RELIEF

Stress is key for survival, but chronic stress can be detrimental to our health. Experiencing emotional, psychological or physical stress for weeks or months can weaken the immune system and cause high blood pressure, fatigue, depression, anxiety and even heart disease. Find a way to help reduce stress every day - this may include exercise such as walking or yoga.

Using **meditation, mindfulness and guided visualisations** can also help to relieve anxiety, overwhelm and stress. These exercises help focus and calm the MIND.

**Practice breathing exercises:** Deep diaphragmatic breathing helps us to relax, become quiet and heal. This can be done anywhere and only needs a few minutes of time. Breathing helps calm the BODY.

## QUESTIONS:

1. Why may you suggest that a customer considers exercising, what would be the benefits?
2. What kind of exercises would be beneficial?
3. What tips would you give to a customer who is struggling to sleep?
4. Name 2 reasons why it is beneficial to get out into nature?
5. Why may a customer want to limit their exposure to radiation from electronic devices?
6. Name 3 ways to reduce stress?

[illegible]

# 5

## MODULE FIVE

### SUPPLEMENTS & NUTRICEUTICALS

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*"The food you eat can either be the safest and most powerful form of medicine or the slowest form of poison"*

- Ann Wigmore

**Natural Health Products** are defined as vitamins, minerals, herbal remedies, homeopathic medicine, traditional medicine, probiotics and other natural products.

**Natural medicine** can be defined as any system of medicine that compliments and enhances the body's natural capacity to heal by restoring balance without the use of synthetic drugs.

## WHAT ARE SUPPLEMENTS AND NUTRACEUTICALS

### & WHY DO WE NEED THEM?

Humans depend on an array of different nutrients in order to thrive and be able to prevent various diseases. Nutritional supplements won't compensate for a bad diet. However, even with a healthy and balanced diet there may be circumstances that call for the use of supplements:

- Depleted soil typically lacks important minerals and trace elements
- Foods that have been genetically modified, sprayed with pesticides, processed, packaged and trucked long distances or kept in cold storage have lost a lot of their nutritional value
- The process of refining, boiling, frying, and exposing food to other types of preparation reduces its content of many different nutrients
- In many parts of the world, we are unable to synthesise enough vitamin D from the sunlight during the winter period
- Environmental pollution, stimulant abuse, and the use of chronic medicines challenges the body's immunity and cleansing organs
- Stress, pregnancy, lactation, age, disease, and high-performance sports activities often increase the need for a number of different nutrients

**A supplement** is a product that contains one or more of the following dietary ingredients: A vitamin, a mineral, an amino acid, a medical herb or other botanical. A supplement is a dietary substance used to supplement the diet by increasing the total daily nutritional intake.

**Nutraceuticals** are supplements which are used for health purposes other than nutrition. A nutraceutical product may be defined as a substance which may be used to improve health, delay the ageing process, prevent chronic diseases, increase life expectancy, or support the structure or function of the body.

#### **Multivitamin & Mineral**

For the human body to complete all the tasks it has in a normal day, it must be given a wide and complex variety of vital nutrients. Ideally these should come from a wholefoods diet containing a wide variety of fresh foods. A high-potency multivitamin/multi-mineral formula may help ensure the presence of the essential nutrients necessary for the thousands of metabolic reactions in the body. Multi-vitamins are convenient and easy to use.

#### **Vitamin and Mineral Toxicity**

A condition in which a person develops symptoms as side effects from taking large doses of vitamins, minerals or fat soluble vitamins which may build up to dangerous levels in the body. It is always advisable to first obtain vitamins and minerals from healthy food sources, as it is almost impossible to overdose with food. Only take therapeutic doses under supervision of a physician. Please see the vitamins and minerals chart for toxicity symptoms, and the daily recommended intake.

## RELIABLE NUTRACEUTICAL AND SUPPLEMENT BRANDS

Wellness Warehouse offers a large range of different supplement brands. Some brands are only available at Wellness Warehouse or Health Shops. These brands are more exclusive and generally are from companies that have a good reputation.

These companies ensure that the raw material used in the supplements are tested for efficacy and quality. They manufacture in pharmaceutical grade facilities. They also offer the supplements in optimal doses so that the benefits can be felt.

### Brands to recommend to customers:

- Wellness Warehouse Own Brand
- Solgar
- Metagenics
- Good Health
- Viridian
- Vibrant Health
- Future Health
- Sfera



## READING A SUPPLEMENT LABEL

### • Delivery Methods:

- **Tablets:** an accurately measured dosage of the powdered active ingredient in a convenient portable compressed form.

- **Capsules:** single-piece gel encapsulation (soft capsules or soft gels)- Primarily used for oils and for active ingredients that are dissolved or suspended in oil. Two-piece gel encapsulation (hard capsules) - Usually contain dry, powdered ingredients or miniature pellets. Capsules may be made from gelatin (bovine or porcine) or vegetarian (pine or poplar cellulose).

• **Recommended Daily Value** is the amount needed to prevent disease. If the % Daily Value listed is 75%, it means one serving supplies 75% of the recommended Daily Value for that nutrient. Sometimes the Daily Value of supplements can be over 100%, this is because this may be the amount that provides the most benefit.

• **Serving size** is the amount that is needed to be consumed in a day to reach the Daily Value. The ingredient list may be for the serving size and not per tablet/capsule, make sure to compare like with like.

### • Units of measure:

- 1g = 1000mg
- 1 mg = 1000mcg
- 1 mcg = 1μ
- IU = International Units used for fat soluble vitamins A, D, E and K

• **Form of nutrient** - we like to see the detail of what form the vitamin or mineral is in. Active forms of vitamins are ready for the body to use without having to convert into the active form.

• **Other ingredients** - these are ingredients that help bind the tablet or fill the capsule, help the ingredients flow better or preserve the ingredients.

• **Directions and dosage recommendations** - make sure to follow the directions and dosage recommendations for the product, never exceed the dosage instructions.



- **Contra-indications** - draw the customer's attention to any contra-indications as indicated on the label.
- **Storage instructions** - make sure to remind the customer how to store the product if it needs to be refrigerated. All supplements should be stored in a cool dark place. Cotton wool should be removed from the bottle as it can attract moisture.
- **Slow Release** - Slow or sustained release supplements are designed to promote delayed and continuous absorption. They may be encapsulated in a specific way, have a hard outer coating or combined with active ingredients that form a gel like material that is not dissolved upon digestion.
- **Liposomal vitamins and minerals** - Water soluble vitamins B and C, as well as electrolytes such as magnesium that are bound to fatty acids to enable them to stay in the body for longer and achieve a higher absorption rate. Generally better absorbed than slow or sustained release products.

Some customers due to their religion or personal choices cannot take supplements that contain certain ingredients. Please be mindful of finding the most suitable supplements for these customers. The most common concerns are:

- **Vegans/Vegetarians**
  - Avoid gelatin capsules from bovine or porcine sources
  - Avoid animal based digestive enzymes such as ox bile, rennet, lipase and pepsin
  - Avoid animal based fillers such as Magnesium Stearate and Caprylic Acid
  - Avoid vitamin D3 made from lanolin
  - Avoid supplements containing honey or propolis
- **Halaal/Kosher**
  - Supplements must be certified kosher or halaal with certification logo
  - Avoid alcohol in herbal tinctures, cough syrups or products extracted with alcohol (vanilla extract)
  - Gelatine capsules made from porcine sources



# 1. VITAMINS

Vitamins are organic compounds that are needed in small quantities to sustain life. Most vitamins need to come from food because the human body either does not produce enough of them, or it does not produce any at all. Vitamins are vulnerable to heat, light, and chemical agents, so food preparation, processing, and storing them must be appropriate to preserve them.

## FAT-SOLUBLE VITAMINS

Fat-soluble vitamins are stored in the fatty tissues of the body and the liver. Vitamins A, D, E, and K are fat-soluble. These are easier to store than water-soluble vitamins, and they can stay in the body as reserves for days, and sometimes months. Fat-soluble vitamins are absorbed through the intestinal tract with the help of fats, or lipids. Vitamin toxicity is possible because of the build-up of these vitamins in the body.

### Vitamin A

- Immune system
- Tissue repair and healing
- Respiratory health
- Skin health as well as acne
- Protects against eye degeneration



### Beta Carotene

The most abundant carotenoid which can be converted into the active Vitamin A form by the body.

- Anti-oxidant and cell protection
- Immune system
- Tissue repair and healing
- Respiratory health
- Skin health as well as acne
- Protects against eye degeneration

### Vitamin D3

The body can make Vitamin D from sunlight but we often do not get enough sun exposure and darker skins make it harder to absorb adequate amounts.

- Immune function and disease prevention
- Regulates minerals involved in bone health
- Hormone regulation
- Concentration, memory and mood
- Anxiety and insomnia



### Vitamin E

Use natural forms which are D-alpha and mixed forms tocopherol and tocotrienols.

- Important anti-oxidant that should be used with Vitamin C
- Good for cardiovascular eg high blood pressure and circulatory support
- Protects brain and nervous system.
- Useful for skin healing and skin health eg acne, scars, eczema and sunburn
- Useful for female health and male infertility



### Vitamin K1

- Helps for blood clotting so not to be used by those on blood thinning medication.



### Vitamin K2

- Usually used together with Vitamin D3 for helping to ensure good bone health.



# WATER-SOLUBLE VITAMINS

Water-soluble vitamins do not stay in the body for long. The body cannot store them, and they are soon excreted in urine. Because of this, water-soluble vitamins need to be replaced more often than fat-soluble ones. Vitamin C and all the B vitamins are water soluble.

## B-VITAMINS

It is advisable to recommend taking the B-vitamins together as a complex.

### Vitamin B1 - Thiamin

- Production of energy, metabolism of carbohydrates and nerve function
- Poor appetite and weight loss
- Helps metabolise alcohol in liver

### Vitamin B2 - Riboflavin

- Used for energy production and manufacture of amino acids and fatty acids
- Useful for eye health eg cataracts
- Migraines
- Excess is excreted and makes urine bright yellow which is harmless

### Vitamin B3 - niacin, nicotinic acid, nicotinamide and inositol hexanicotinate.

- For cardiovascular health: cholesterol, triglycerides, lipoprotein A and circulation (use Niacin and inositol hexanicotinate)
- Also important for blood sugar regulation
- Releases stored histamine (flushing) where there are signs of histamine deficiency
- Memory loss

### Vitamin B5 - Pantothenic acid

- Fatigue - needed for the production of energy
- Important for physical endurance
- Chronic fatigue
- Adrenal gland support where there is adrenal exhaustion
- Cholesterol and triglycerides
- Memory loss and depression
- Burning feet syndrome

### Vitamin B6 - pyridoxine, pyridoxal and pyridoxal-5-phosphate P5P (active form)

- PMS
- Cardiovascular disease
- Blood pressure and cholesterol
- ADHA
- Depression and anxiety
- Important for methylation and reducing homocysteine
- Pregnancy related nausea

### Vitamin B12 - cobalamin and methylcobalamin (active form)

Absorbed better through the mouth, sublingual tablets, drops or spray and through the skin as a patch.

- Important for vegans and vegetarians
- Older people more likely to be deficient
- Needed for manufacture of red blood cells, deficiency can cause pernicious anaemia
- Fatigue
- Important for methylation and homocysteine and cardiovascular health
- Important for nervous system as well as mental and emotional health (depression, insomnia and memory)

### Biotin

- Hair, thinning hair, dandruff and dermatitis on scalp (cradle cap)
- Nail health as well as brittle nails
- For diabetics needed for proper glucose metabolism and preventing diabetic neuropathy



## Folate

Folate (5-methyltetrahydrofolate 5-MTHF) is the active form, also available as folic acid and folacin.

- Foetal development, pregnancy
- Mood, mental clarity and memory
- Methylation of homocysteine
- Nervous system health
- Restless leg syndrome

## Vitamin C

Found as ascorbic acid, buffered vitamin C, Ester C or Liposomal (better absorbed and stays in the body for longer)

- Stress and adrenal fatigue, together with vitamin B
- Antioxidant and cell protection
- Immune support and recurring infections
- Allergic reactions eg hay fever, eczema, asthma
- Skin health, skin ulcers, poor wound healing and gum disease
- Degenerative eye disease
- Cardiovascular disease eg high blood pressure
- Detoxification side effects
- Collagen manufacture and synthesis
- Helps with absorption of Iron
- Chronic inflammation



## 2. MINERALS

Minerals are inorganic elements that occur naturally in the crust of the Earth and always retain their chemical identity. Once minerals enter the body, they remain there until excreted, they cannot be changed into anything else. Minerals cannot be destroyed by heat, air, acid, or mixing.

**Macro minerals** are needed in dosages greater than 100mg a day. These include: Calcium, magnesium, phosphorus, potassium, sodium, chloride, and sulphur.

**Trace minerals** belong to a large group of nutrients which we only need in microscopic (trace) amounts. Despite the extremely limited amounts of these nutrients in the human body, they are of vital importance. These include: Selenium, chromium, iron, iodine, copper, manganese, silica and zinc

### Foodstate minerals

When we ingest minerals from the food we eat such as vegetables and animal food sources, the compounds are in their organic form and easy for the body to absorb.

### Elemental Value

For a mineral to be absorbed into the body it has to be bound to a larger molecule. These larger molecules can be inorganic compounds such as carbonates, oxides, phosphates and sulphates or organic compounds such as citrates, picolates, glycates and malates. These organic bonds are easier to break in the intestine and the mineral can now bind to a protein (amino acid) and be absorbed. The elemental value refers to the actual amount or weight of the mineral present, and not the weight of the mineral plus the bound molecule. It gives more accurate information on the label of how much of the mineral is actually supplied.

### Chelated Minerals

A mineral may be chelated when it is bound to an amino acid, it is then called a chelated mineral. The mineral does not have to be separated from its carrier with the use of stomach acid. It is therefore easier for the body to absorb and utilise. Chelated mineral forms: glycinate, bisglycinate and taurate.

## Boron

- Menopause
- Osteoporosis and bone density
- Joint health
- Concentration and brain function
- Building muscle and burning fat

## Calcium - best taken in a complex with magnesium

- Bone density and healing fractures
- Dental health
- Leg cramps and muscle spasms - with magnesium
- Found in alkaline powders for balancing pH
- Nervous tension and irritability - with magnesium



## Chromium

- Blood sugar control
- Diabetes and insulin resistance
- Cravings for sugar and refined carbohydrates
- High cholesterol and triglycerides
- Weight control
- Sports Nutrition - helps amino acids get into muscles



## Copper

- Fatigue and weakness
- Weak and brittle bones
- Weak immune system
- Vein health eg spider veins, varicose veins and aneurysms.
- Vision loss and cataracts
- Premature grey hair
- Problems with memory and learning
- Essential to help absorb iron
- Zinc can interfere with the absorption of copper as these nutrients compete for the same absorption sites



## Iodine

- Thyroid function - useful for low thyroid
- Weight control
- Growth and development during pregnancy and breast feeding

## Iron

- Anaemia - pregnancy or heavy periods
- Pale children with no energy or appetite
- Fatigue and general weakness
- Vegans and vegetarians
- Restless leg syndrome
- Should have regular blood tests and take iron in non-constipating form and with Vitamin C

## Magnesium

Can be taken as a supplement and also used in a transdermal form as a spray

- Calcium regulation and bone health
- Muscle cramps, stiff muscles and spasms
- Period pains or cramps
- Fibromyalgia
- Insomnia
- Migraines and headaches when due to tension
- Constipation
- High blood pressure - take with Potassium
- Cardiovascular support
- Anxiety
- Stress
- Blood sugar control



## Manganese

- Arthritis, sprains and strains, important for connection tissue health and repair
- Metabolism of blood sugar and lipids

## Phosphorus

- Strong teeth and bones
- Fatigue and physical weakness
- Part of cell membrane - phospholipid formation
- Cell integrity

## Potassium

- High blood pressure together with magnesium
- Fluid retention and sodium regulation
- Dehydration, vomiting and diarrhoea
- Fatigue and exhaustion
- Muscle weakness
- Stress depletes potassium

## Selenium

- Antioxidant nutrient
- Protects against free radical damage and the ageing process
- Liver protection and detoxification
- Thyroid function
- Prostate health
- Fertility
- Pregnancy for growth and development
- Immune support

## Silica

- Bone health and osteoporosis
- Ageing skin and wrinkles
- Weak thin hair
- Weak brittle nails
- Joint health

## Sulphur

- Sold as MSM (methylsulfonylmethane)
- Improves immune function
- Lowers chronic inflammation
- Restores bodily tissue and improves skin conditions
- Treats osteo-arthritis and joint pain
- Improves digestive problems
- Supports respiratory health
- Lowers muscle pain/spasms and improves exercise recovery

## Zinc

- Antioxidant nutrient
- Skin healing, acne, eczema, psoriasis
- Immune support and viral infections
- Female hormonal health, eg growth and repair in pregnancy, infertility
- Male hormonal health, eg prostate, low sperm count, infertility
- Degenerative eye conditions, eg cataracts and macular degeneration
- Mental health, eg ADHD, learning difficulties, dementia
- Digestive health eg leaky gut, Crohn's disease and celiac disease
- Excessive intake of zinc may suppress copper and iron absorption



## WATER SOLUBLE VITAMINS

mg = milligrams

mcg = micrograms (equal to µ)

UL = Tolerable Upper Intake Level for adults  
IU = International Units

| Vitamin Name                                      | Primary Functions                                                                                                                                                           | Recommended Intake                                                                                                                                                                                                                                                                        | Reliable Food Sources                                                                                                                                | Deficiency Symptoms                                                                                                                                                                                       | Toxicity Symptoms                                                                                                                                                                                                                                                             | Interactions with drugs or other vitamins                     | Recommend for                                                                                                                                      |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Thiamin (Vitamin B1)                              | Protein and carbohydrate metabolism into energy, heart and nerve function                                                                                                   | <p>Infants:<br/>0-6 mnths = 0.2 mg/day<br/>7-12 mnths, 0.3 mg/day</p> <p>Children:<br/>1-3 yrs = 0.5 mg/day<br/>4-8 yrs = 0.6 mg/day<br/>9-13 yrs = 0.9 mg/day</p> <p>Adults:<br/>Men = 1.2 mg/day<br/>Women = 1.1 mg/day<br/>Pregnant and breast Feeding:<br/>1.4-1.5 mg/day</p>         | Nutritional yeast, brewer's yeast, Spirulina, sunflower seeds, macadamia nuts, lentils, beans and peas, wholegrains, brown rice, blackstrap molasses | Beriberi (loss of appetite, muscle wasting, enlarged heart, pain in the limbs, shortness of breath and swollen feet or legs); fatigue, apathy, decreased memory, confusion, irritability, muscle weakness | None known                                                                                                                                                                                                                                                                    |                                                               | Anorexia or rapid weight loss, poor appetite, nerve damage and inflammation, alcoholism, cataracts                                                 |
| Riboflavin (Vitamin B2)                           | Carbohydrate and fat metabolism into energy                                                                                                                                 | <p>Infants:<br/>0-6 mnths = 0.3 mg/day<br/>7-12 mnths = 0.4 mg/day</p> <p>Children:<br/>1-3 yrs = 0.5 mg/day<br/>4-8 yrs = 0.6 mg/day<br/>9-13 yrs = 0.9 mg/day</p> <p>Adults:<br/>Males 14 yrs + = 1.3 mg/day<br/>Females 14 - 18 yrs = 1.0 mg/day<br/>Females 19 yrs + = 1.1 mg/day</p> | Liver & organ meats from grassfed animals, brewer's yeast, wild rice, whole grains, Spirulina, organic feta and goats cheese, almonds, eggs, tahini  | Sores in the mouth; swollen mouth and throat; red, scaling rash on the scalp, face, ears, and torso; anemia,                                                                                              | None known                                                                                                                                                                                                                                                                    | Vitamin B2 is required to activate Vitamin B6 and folic acid. | Anemia, cataracts, migraines, fatigue                                                                                                              |
| Niacin, nicotinamide, nicotinic acid (Vitamin B3) | Carbohydrate and fat metabolism for energy, skin health, brain function, cardiovascular health                                                                              | <p>Children:<br/>Between 2-16 mg/day, depending on age</p> <p>Adults:<br/>Men = 16 mg/day<br/>Women = 14 mg/day<br/>Pregnant and breastfeeding = 17-18 mg/day<br/>UL = 35 mg/day</p>                                                                                                      | Beef liver, chicken breast, tuna, sunflower seeds, grassfed beef, lamb, wild caught salmon, split green peas, sardines, turkey, mushrooms, tahini    | Pellagra (diarrhoea, dermatitis and dementia); Brain impairment and psychosis- impaired consciousness, cognitive decline (dementia), disorientation, confusion, depression, mania, or paranoia.           | Flushing, increased blood sugar, blurred vision; nausea or vomiting; allergic reactions such as rashes, headaches, dizziness; heart problems or irregular heartbeat. Aggravated gallbladder, liver disease or gout symptoms. Low blood pressure, stomach or intestinal ulcers |                                                               | Cardiovascular disease, high cholesterol, diabetes, blistering skin conditions, headaches, memory loss, Alzheimers                                 |
| Pantothenic Acid (Vitamin B5)                     | Protein, fat and carbohydrate metabolism; cholesterol synthesis and cardiovascular health; improves mental performance, balances hormones, maintains heart & nerve function | <p>Infants:<br/>0-6 mnths = 1.7 mg/day<br/>7-12 mnths = 1.8 mg/day</p> <p>Children:<br/>1-3 yrs = 2 mg/day<br/>4-8 yrs = 3 mg/day<br/>9-13 yrs = 4 mg/day</p> <p>Adults:<br/>Men &amp; Women 14 yrs + = 5 mg/day<br/>Pregnant = 6 mg/day<br/>Breastfeeding = 7 mg/day</p>                 | Beef liver, avos, broccoli, sunflower seeds, duck, portabella mushrooms, eggs, salmon, lentils, yoghurt, broccoli                                    | Extremely rare                                                                                                                                                                                            | Mild intestinal distress                                                                                                                                                                                                                                                      |                                                               | High cholesterol and triglycerides, cardiovascular disease, migraines, memory loss, depression, alcoholism, adrenal fatigue, burning feet syndrome |

## WATER SOLUBLE VITAMINS

| Vitamin Name                                                      | Primary Functions                                                                                                                                                                                                                                                                                                                           | Recommended Intake                                                                                                                                                                                                                           | Reliable Food Sources                                                                                                                                                        | Deficiency Symptoms                                                                                                                                                                                     | Toxicity Symptoms                                                                                                                                                                                                                                                             | Interactions with drugs or other vitamins                                                                                                                                                                                                                                                                                                 | Recommend for                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pyridoxine, pyridoxal, pyridoxamine (Vitamin B6)                  | Carbohydrate and amino acid metabolism, helps with manufacture of mood influencing hormones norepinephrine and serotonin, as well as body clock regulating hormone melatonin, helps control homocysteine, natural pain treatment, supports brain function, is needed for B12 absorption and manufacture of immune cells and red blood cells | Men & women aged 19 to 50 = 1.3 mg/day, Men aged > 50 = 1.7 mg/day, Women aged > 50 = 1.5 mg/day, UL = 100 mg/day                                                                                                                            | Turkey breast, grassfed beef, pistachios, tuna, avocado, chicken breast, blackstrap molasses, sunflower seeds, sesame seeds.                                                 | Anemia, dermatitis, convulsions, elevated homocysteine levels; changes in mood, such as irritability, anxiety and depression, confusion; muscle pains, low energy or fatigue, worsening of PMS symptoms | Nerve damage, skin lesion                                                                                                                                                                                                                                                     | Some medications that interact with vitamin B6 include drugs used for treating Parkinson's and Alzheimer's disease, anemia, seizures, tuberculosis, heart disease, high blood pressure, rheumatoid arthritis, asthma, some antibiotics, chemotherapy drugs and antidepressant medications<br>Magnesium + Vitamin B6 is excellent for ADHD | ADHD rheumatoid arthritis, PMS symptoms, for healthy blood vessels, to lower homocysteine, heart disease, blood pressure, cholesterol                                                                                                                                       |
| Biotin (Vitamin B7)                                               | Carbohydrate, fat and protein metabolism, beneficial for hair, skin and nails                                                                                                                                                                                                                                                               | Infants:<br>0-7 mnths = 5 mcg/day<br>7 mnths - 3 yrs = 6-8 mcg/day<br>Children:<br>4-13 yrs = 12-20 mcg/day<br>14-19 yrs = 25 mcg/day<br>Adults:<br>Men & Women 19 yrs + = 30 mcg/day<br>Pregnant = 30 mcg/day<br>Breastfeeding = 35 mcg/day | Organ meats, eggs,                                                                                                                                                           | Although very rare: fatigue, loss of appetite, dry scaly skin, hair loss, magenta coloured and swollen tongue, cracked mouth corners, dry eyes, depression, insomnia                                    | None known                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                           | Pregnancy, long-term use of certain anti-seizure medications, alcoholism<br>consuming lots of raw egg whites, smoking, prolonged antibiotic use, intestinal malabsorption issues or serious digestive disorders like Crohn's disease, celiac disease, or leaky gut syndrome |
| Folic acid (natural form)<br>Folate (synthetic form)              | It aids in cell division and helps make new cells by copying and creating DNA. Helps the body use vitamin B12 as well as certain amino acids. Involved in metabolism of homocysteine                                                                                                                                                        | Adults:<br>Men & women = 400 mcg/day<br>Pregnant = 600 mcg/day<br>Breastfeeding = 500 mcg/day<br>UL = 1000 mcg/day                                                                                                                           | Fortified cereals, enriched breads and grains, legumes (lentils, chickpeas, beans), asparagus, avos, broccoli, brussels sprouts, leafy greens (spinach, lettuce), Beef liver | Fatigue, mouth sores, anemia, neural tube defects in developing fetus, elevated homocysteine levels                                                                                                     | Masks symptoms of Vitamin B12 deficiency, specifically signs of nerve damage, cramps, diarrhea, confusion and skin reactions.<br>Also epilepsy, changes in sex drive, difficulty sleeping and mood changes. Possible increased risk of prostate cancer and colorectal tumors. | Activates B12                                                                                                                                                                                                                                                                                                                             | Restless legs syndrome, pregnancy, cognitive function, strong bones, heart health                                                                                                                                                                                           |
| Methylcobalamin (natural form)<br>Cyanocobalamin (synthetic form) | Assists with formation of blood, required for healthy nervous system function, involved in metabolism of homocysteine                                                                                                                                                                                                                       | Infants:<br>0-6 mnths = 0.4 mcg/day<br>7-12 mnths = 0.5 mcg/day<br>Toddlers:<br>1-3 yrs = 0.9 mcg/day<br>Children:<br>4-8 yrs = 1.2 mcg/day<br>9-13 yrs = 1.8 mcg/day<br>Adults: 2.4 mcg/day<br>Breastfeeding: 2.8 mcg/day                   | Grassfed beef liver, sardines, mackerel, lamb, wild caught salmon, fortified nutritional yeast, feta cheese, cottage cheese, eggs                                            | Anemia, tingling and numbness of extremities, nerve damage, memory loss, disorientation, depression, dementia, elevated homocysteine levels                                                             | None known                                                                                                                                                                                                                                                                    | Medications that are known to decrease the levels of vitamin B12 in the body include: anti-seizure medications, chemotherapy medications, Glucophage, proton pump inhibitors, antibiotics & Metformin                                                                                                                                     | Vegetarians & vegans, smokers (nicotine blocks absorption), alcoholics, those with anemia, and anyone with a digestive disorder like celiac or Crohn's disease.                                                                                                             |

## WATER SOLUBLE VITAMINS

| Vitamin Name              | Primary Functions                                                                                                                                                                                                                                      | Recommended Intake                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Reliable Food Sources                                                                                                                                                                                                                                              | Deficiency Symptoms                                                                                                                                                                                                                                                                                                                                                                                               | Toxicity Symptoms                                                                                    | Interactions with drugs or other vitamins | Recommend for                                                                                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ascorbic Acid (Vitamin C) | Anti-oxidant, assist with collagen synthesis, repair and growth of body tissues, wound healing, teeth and bone repair and maintenance, enhances immune function, assists in synthesis of hormones, neurotransmitters and DNA, enhances iron absorption | Adults:<br>Men and women 14 yrs+ = 2.4 mcg/day<br>Pregnant = 2.6 mcg/day<br>Breastfeeding = 2.8 mcg/day<br>Infants:<br>0 – 6 mnths = 40 mg/day<br>7 – 12 mnths = 50 mg/day<br>Children:<br>1 – 3 yrs = 15 mg/day<br>4 – 8 yrs = 25 mg/day<br>9-13 years = 45 mg/day<br>Adolescents:<br>Girls 14 – 18 yrs = 65 mg/day<br>Pregnant teens = 80 mg/day<br>Breastfeeding teens = 115 mg/day<br>Boys 14 – 18 yrs = 75 mg/day<br>Adults:<br>Men 19+ yrs = 90 mg/day<br>Women 19 + yrs = 75 mg/day<br>Pregnant = 85 mg/day<br>Breastfeeding = 120 mg/day<br>Smokers, or those who are exposed to second hand smoke, should increase their daily amount of vitamin C by 35 mg/day<br>UL = 2000mg | Superfoods such as acerola cherries, goji berries, baobab and camu camu; fruits and veg such as guavas, red currants, red and green sweet peppers, kiwis, citrus fruits, strawberries, papayas, pineapples, mangoes, broccoli, kale, parsley, cabbage, cauliflower | Scurvy, easy bruising; swollen or bleeding gums; Gingivitis (inflammation of the gums); slow wound healing, dry and splitting hair, dry red spots on the skin; rough, dry, scaly skin; nosebleeds, weakened immune system; digestive disorders like leaky gut and autoimmune disease. Possible weight gain because of slowed metabolism. Swollen and painful joints; bone pain and fractures; depression, anemia. | Nausea and diarrhea, nose bleeds, increased formation of kidney stones in people with kidney disease | Enhances iron absorption                  | For stress, depression, recurrent infections, immunity support, high blood pressure, gallbladder disease, stroke, certain cancers, anemia, atherosclerosis |

## FAT SOLUBLE VITAMINS

mg = milligrams  
mcg = micrograms (equal to µ)

UL = Tolerable Upper Intake Level for adults  
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| Vitamin Name                                                | Primary Functions                                                                                                                                                                                                                                                                                                 | Recommended Intake                                                                                                                                                                                                                                                                                                                                                        | Reliable Food Sources                                                                                                                                                                                                                                                                                                                                        | Deficiency Symptoms                                                                                                                                                                                                                                                   | Toxicity Symptoms                                                                                                                                                                                                                   | Interactions with drugs or other vitamins                                                                                                                                                                                                                                                                                                  | Recommend for                                                                                                                                                                                                                                  |
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| Retinol, retinal, retinoic acid, beta carotene (Vitamin A)  | An important anti-oxidant, protects against eye disease, important for fertility, contributes to healthy bones and immune system, assist with protection of the mucous membranes of the digestive, respiratory, and urinary tracts against infection, important for skin health, helps with reproduction of cells | Children:<br>1-3 yrs = 300 mcg/day<br>4-8 yrs = 400 mcg/day<br>9-13 yrs = 600 mcg/day<br>Adults:<br>Men = 900 mcg/day (3000 IU)<br>Women = 700 mcg/day (2333 IU)<br>Pregnant = 750-770 mcg/day<br>Breastfeeding: 1,200-1,300 mcg/day<br>UL = 3000 mcg/day (10 000 IU)                                                                                                     | Retinoids: fish, beef and chicken liver, egg yolks, milk<br>Beta carotene: green leafy veg (spinach, kale) yellow & orange fruit & veg (carrots, mangoes, apricots, melons, pumpkin, sweet potatoes)                                                                                                                                                         | Night blindness; abnormal dryness of tear ducts and eyes; dry, scaling and thickening skin; impaired growth, immunity and reproductive function. Alcoholics and persons with digestive illnesses are most at risk of deficiency because of malabsorption of nutrients | Fatigue; bone and joint pain; spontaneous abortion and birth defects of fetuses in pregnant women; nausea and diarrhea; liver damage, nervous system damage, blurred vision, hair loss, skin disorders, haemorrhage, coma and death | Vitamin A supplements may interact with some birth control pills, blood thinners (like Warfarin, Coumadin), acne medicines (like Accutane). Zinc and vitamin D deficiencies might worsen vitamin A deficiency. People who drink heavily or have kidney or liver disease should not take vitamin A supplements without talking to a doctor. | Acne; wound healing; lung-, prostate-, breast-, ovarian-, bladder-, oral-, and skin cancers, respiratory infections, food allergies, recurrent infections; childhood illnesses such as measles, dry eyes, prevention of age-related blindness. |
| Vitamin D3 (Cholecalciferol)<br>Vitamin D2 (Ergocalciferol) | Regulates absorption of calcium and maintains bone health; facilitates hormone regulation and improves mood; helps with concentration, learning and memory; enhances immune system;                                                                                                                               | Babies to 1 yr: 400 IU/day<br>Children:<br>1-8 yrs: 600 IU (15 mcg/day)<br>Older children & Adults: 9 to 70 yrs= 600 IU/day (15 mcg/day)<br>older than 70 yrs = 800 IU/day (20 mcg/day). Pregnant and breast-feeding: 600 IU (15 mcg/day)<br>UL = 4000 IU/day (100 mcg)                                                                                                   | Vitamin D3: fatty fish such as salmon, mackerel and tuna; cod liver oil; full cream, raw and grass fed milk, cream and butter; 10 to 20 minutes of time in the sun daily (between 1000 and 10 000 IU); egg yolk and liver, lichens.<br><br>Vitamin D2: plants such as maitake and portabella mushrooms grown in sunlight or under UV light; fortified foods. | Rickets (weak & soft bones) in children, osteomalacia (softening of the bones) and/or osteoporosis (fragile bones) in adults                                                                                                                                          | Hypercalcemia (too much calcium in the blood) which can cause nausea and vomiting, weakness, and frequent urination. Symptoms might progress to bone pain and kidney problems, such as the formation of calcium stones.             | Vitamin K2 should always be given with vitamin D3; Supplementing with very high doses of vitamin A is not recommended if you have a known vitamin D deficiency                                                                                                                                                                             | Osteoporosis, heart disease, high blood pressure, cancer, autoimmune diseases, depression, anxiety, PMS, insomnia, arthritis, diabetes, asthma, multiple sclerosis, chronic pain, psoriasis, fibromyalgia, ADHD, autism                        |
| Tocopherol (Vitamin E)                                      | A powerful anti-oxidant and anti-inflammatory, has anti-cancer and anti-tumor abilities, cholesterol-lowering effects. Prevents diseases of the heart and blood vessels, repairs skin damage, balances hormones, protects against cognitive decline, boosts stamina and vitality                                  | Infants:<br>0-6 months: 2.0 mcg/day<br>7-12 months: 2.5 mcg/day<br>Children:<br>1-3 yrs = 6 mg/day (9 IU)<br>4-8 yrs = 7 mg/day (10.5 IU)<br>9-13 yrs = 11 mg/day (16.5 I)<br>Adults:<br>Men: 14+ yrs = 15 mg/day (22.5 IU)<br>Women: 14+ yrs = 15 mg/day (22.5 IU)<br>Pregnant = 15 mg/day (22.5 IU)<br>Breastfeeding = 19 mg/day (28.5 IU)<br>UL = 1000mg/day (1500 IU) | Vitamin E is found only in plant foods, including certain oils (coconut, palm, olive, rice bran and wheat germ), seeds (sunflower seeds), nuts (peanuts & peanut butter, almonds, pecans and walnuts), fruits (tomato, avo, papaja), veg (spinach, sweet potatoes, butternut), grains (oats, rye and barley)                                                 | Hemolytic anemia (red blood cell breakage), impairment of nerve, muscle and immune function                                                                                                                                                                           | Rare                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                            | Cholesterol, high blood pressure, heart disease, hardening and calcification of arteries, acne, scars, ecsema, sunburn, PMS, Alzheimers, dementia, pregnancy, cancer                                                                           |

## FAT SOLUBLE VITAMINS

| Vitamin Name                                           | Primary Functions                                                                                                                                                                               | Recommended Intake                                                                                                                                                                                                                                 | Reliable Food Sources                                                                                                                                                                                                                                                                                                                                                                                                                  | Deficiency Symptoms                                                                                                                                                                                                                                                                                                                                                                           | Toxicity Symptoms | Interactions with drugs or other vitamins                                                                                                                                                                                                                                                                                                                                                                  | Recommend for                                                         |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Phylloquinone (Vitamin K1)<br>Menaquinone (Vitamin K2) | Vitamin K1 serves as a co-enzyme during production of proteins that assist in blood coagulation.<br>Vitamin K2 assists with heart and brain health, building strong bones and preventing cancer | Infants:<br>0–6 months: 2.0 mcg/day<br>7–12 months: 2.5 mcg/day<br>Children:<br>1 – 3 years: 30 mcg/day<br>4 – 8 years: 55 mcg/day<br>9 – 13 years: 60 mcg/day<br>Adults:<br>Men & Women:<br>14 – 18 years: 75 mcg/day<br>19 and older: 90 mcg/day | Vitamin K1: Leafy greens (kale, spinach, dandelion greens, mustard greens, swiss chard, spring onions); Cruciferous veg (broccoli, brussel sprouts, cabbage, cauliflower); Other green veg (asparagus, kelp, cucumbers)<br><br>Vitamin K2: Fermented foods (sauerkraut, kimchi, kombucha, kefir); Grass-fed organic animal products (i.e. eggs, butter, dairy); Certain cheeses (Brie and Gouda). Only reliable vegan source is natto. | Deficiency symptoms include impaired blood clotting; easy bruising; excessive bleeding from wounds, punctures, and injection or surgical sites; heavy menstrual periods; bleeding from the gastrointestinal (GI) tract; blood in the urine and/or stool.<br>Taking blood thinners, dealing with long term hemodialysis, and suffering from a serious burn can lead to a vitamin K deficiency. | None known        | Never give K1 to patients on blood thinning medications, or who have had a stroke or heart attack.<br><br>Always give vitamin K2 together with vitamin D3. Vitamin D3 mobilises calcium into the blood stream and K2 directs calcium to the bones and teeth, and away from soft tissues (heart muscle, arteries and organs) where calcium will harden the tissues. K2 is easy to obtain from food sources. | Always give with Vitamin D3, heart health, bone density, PMS symptoms |

## MINERALS

mg = milligrams

mcg = micrograms (equal to  $\mu$ )

UL = Tolerable Upper Intake Level for adults  
IU = International Units

| Vitamin Name                                 | Primary Functions                                                                                                                                                                                                                                                                                                                                                       | Recommended Intake                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Reliable Food Sources                                                                                                                                          | Deficiency Symptoms                                                                                                                                                                                                                           | Toxicity Symptoms                                                                                                                                                                                                           | Interactions with drugs or other vitamins                                                                                                                                                                                                                                                                                                                                          | Recommend for                                                                                                                  |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Calcium carbonate, Calcium citrate (Calcium) | Important for growth and maintenance of strong teeth and bones, helps in preventing osteoporosis. Helps with muscle, nerve and heart function. Involved in the release of neurotransmitters in the brain that control muscle movement and nerve signaling. Helps with the control and release of glucose (sugar) in the blood stream, and optimizing glucose metabolism | <p>Infants:</p> <ul style="list-style-type: none"> <li>0–6 mnths = 200 mg/day</li> <li>7–12 mnths = 260 mg/day</li> </ul> <p>Children:</p> <ul style="list-style-type: none"> <li>1–3 yrs = 700 mg/day</li> <li>4–8 yrs = 1,000 mg/day</li> <li>9–13 yrs = 1,300 mg/day</li> </ul> <p>Adults:</p> <ul style="list-style-type: none"> <li>14–18 yrs = 1,300 mg/day</li> <li>19–50 yrs = 1,000 mg/day</li> <li>51–70 yrs = 1,000 mg/day (men) &amp; 1200 mg/day (women)</li> <li>71+ yrs 1,200 mg</li> </ul> <p>Pregnant &amp; breastfeeding:</p> <ul style="list-style-type: none"> <li>14–18 yrs: 1300 mg/day</li> <li>19–50 yrs: 1000 mg/day</li> </ul> <p>Safest: No more than 500 mg at one time, taken with food</p>    | <p>Whey protein, kale, sardines, yoghurt &amp; kefir, white beans, kidney beans, tahini, broccoli, watercress, cheese, bok choy, kelp, chia seeds, almonds</p> | <p>Brittle &amp; weak bones, bone fractures, Osteoporosis, problems with proper blood clotting, weakness &amp; fatigue, delays in children's growth &amp; development, heart problems involving blood pressure and abnormal heart rhythms</p> | <p>Nausea, gas, bloating, constipation (especially from calcium carbonate because of its ability to neutralize stomach acid), dry mouth, abdominal pain, irregular heartbeat, confusion, kidney stones, and even death.</p> | <p>Always give in 2:1 ratio with magnesium, and add collagen, Vit D3 and K2. Very high levels of calcium can interact with drugs intended to treat heart disease, diabetes, epilepsy, and other conditions. Calcium can also increase risk for kidney stones and can interfere with the absorption of other important minerals, like iron, magnesium, and phosphorus and zinc.</p> | <p>Healing of fractures &amp; dental health; Osteoporosis; high blood pressure; diabetes, weight loss, GERD or acid reflux</p> |
| Chromium                                     | Important for the hormone insulin to help with regulation of blood sugar levels. Helps with metabolism of fat. Important for brain health and protection against cognitive decline. Maintains eye health. Slows the loss of calcium in the body                                                                                                                         | <p>Infants:</p> <ul style="list-style-type: none"> <li>0 - 6 mnths = 0.2 mcg/day</li> </ul> <p>Children:</p> <ul style="list-style-type: none"> <li>7-12 mnths = 5.5 mcg/day</li> <li>1-3 yrs = 11 mcg/day</li> <li>4-8 yrs = 15 mcg/day</li> <li>9-13 yrs = 25 mcg/day (boys) &amp; 21 mcg/day (girls)</li> </ul> <p>Teens:</p> <ul style="list-style-type: none"> <li>14-18 yrs = 35 mcg/day (boys) &amp; 24 mcg/day (girls)</li> </ul> <p>Adults:</p> <ul style="list-style-type: none"> <li>19-50 yrs = 35 mcg/day (men) &amp; 25 mcg/day (women)</li> <li>Pregnant: 30 mcg/day</li> <li>Breastfeeding: 35 mcg/day</li> </ul> <p>For insulin-resistance or diabetes; 200 micrograms a day as part of a multivitamin</p> | <p>Broccoli, grapes, potatoes, garlic, grass fed beef and free range turkey, green beans, red wine, apples, bananas, oranges, basil</p>                        | <p>Increased blood sugar, levels of cholesterol and triglycerides. Increases risk of heart disease and diabetes.</p>                                                                                                                          | <p>Digestive problems, stomach aches, low blood sugar levels (hypoglycemia). Rarely: damage to vital organs like the liver, kidneys and heart and changes in nerve signaling and heart beat</p>                             | <p>None known</p>                                                                                                                                                                                                                                                                                                                                                                  | <p>Diabetes, insulin resistance, weight management, high cholesterol, glaucoma, osteoporosis</p>                               |
| Copper                                       | Bone & connective tissue production; proper immune function; eye health; cognitive health; absorption of iron; plays a role in making red blood cells                                                                                                                                                                                                                   | <p>Adolescents &amp; adults: 900 mcg/day</p> <p>UL for 19 yrs +: 10 000 mcg/day</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Avos, cashews, chickpeas, dark chocolate, mushrooms, salmon, seafood, sesame seeds, sweet potatoes, tofu, water kept in copper vessels</p>                  | <p>Rare: weak bones, joint pain, vitamin B12 deficiency, iron deficiency, anemia</p>                                                                                                                                                          | <p>Cramps, diarrhoea &amp; vomiting (short term); neurological effects (long term)</p>                                                                                                                                      | <p>Essential for iron absorption. Excess zinc intake suppresses copper absorption (be careful of zinc lozenges over long periods of time)</p>                                                                                                                                                                                                                                      | <p>Alzheimer's disease, osteoporosis, iron deficiency, spider veins, varicose veins, aneurysms</p>                             |

## MINERALS

| Vitamin Name                                                          | Primary Functions                                                                                                                                                                                                                                                                                                                             | Recommended Intake                                                                                                                                                                                                       | Reliable Food Sources                                                                                                                                                                                                                                                                                                                                                                                                                 | Deficiency Symptoms                                                                                                                                                                                                                                                                                               | Toxicity Symptoms                                                                                                                                      | Interactions with drugs or other vitamins                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Recommend for                                                                                                                                                                                  |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Potassium iodine (Iodine)                                             | Needed for making thyroid hormones, which are required for normal development and growth, controls metabolism and is required for optimal energy levels                                                                                                                                                                                       | Children:<br>1-8 yrs = 90 mcg/day<br>9-13 yrs = 120 mcg/day<br>Adults:<br>14+ yrs = 150 mcg/day<br>Pregnant & breastfeeding: 290 mcg/day<br>UL = 1100 mcg/day                                                            | White deep water fish, shellfish, kelp, nori, spirulina, garlic, spinach and Swiss chard, yoghurt, raw milk, eggs, cranberries and Himalayan salt.<br>Eating raw vegetables in the Brassica family (cauliflower, broccoli, kale, cabbage, soy, Brussels sprouts) can impact thyroid function because they contain goitrogens. Steaming these cruciferous vegetables until fully cooked before consumption breaks the goitrogens down. | Low thyroid hormone levels (hypothyroidism) results in dry skin, weight gain, fatigue, depression, menstrual problems, recurrent infections, sensitivity to cold, cold hands and feet, brain fog, thinning hair, shortness of breath, constipation, impaired kidney function, muscle weakness and joint stiffness | Rare: burning of the mouth, throat, and stomach; fever, nausea, vomiting, diarrhea, a weak pulse, cyanosis and coma                                    | Tap water contains fluoride and chlorine that inhibit the absorption of iodine. The thyroid gland needs both selenium and iodine to produce adequate levels of thyroid hormones.                                                                                                                                                                                                                                                                                                                                                          | Low thyroid hormone levels (hypothyroid), smokers, pregnant and breastfeeding women                                                                                                            |
| Iron                                                                  | Needed to produce hemoglobin, a type of protein found in red blood cells carries oxygen from the lungs throughout the body to the cells; supports energy levels, as well as healthy development and growth.                                                                                                                                   | Children:<br>4-8 yrs = 10 mg/day<br>9-13 yrs = 8 mg/day<br>Adults:<br>Men 19-50 = 8 mg/day<br>Women 19-50 yrs = 18 mg/day<br>Pregnant = 27 mg/day<br>Breastfeeding: 10 mg/day                                            | Heme iron: Liver, red meat, shellfish, fish and poultry.<br>Non-heme iron: legumes (chickpeas, lentils & beans), seeds, nuts, whole grains, green leafy vegetables and dark molasses. Raw dark chocolate                                                                                                                                                                                                                              | Anemia, fatigue, weakness, pale skin, shortness of breath, abnormal heartbeat, changes in appetite                                                                                                                                                                                                                | Nausea, vomiting, cramps and constipation.<br>Excessive iron from supplements can be toxic to children, causing serious health problems or even death. | Medications for Parkinson's disease, cancer or heart disease may be malabsorbed when also taking iron. Medications such as antacids and proton pump inhibitors which lead to lower stomach acid will inhibit the absorption of iron. Non-heme iron is better absorbed when taken with vitamin C, fructose (whole fruit or honey). Non-heme iron is not well absorbed when eaten together with eggs, foods high in phytates, oxalates, tannins, polyphenols and fibre. Copper is essential for iron absorption, as is enough stomach acid. | Anemia; healthy pregnancy; children who are pale and has no energy or appetite; vegans and vegetarians; restless leg syndrome                                                                  |
| Magnesium Citrate, Magnesium Chelate, Magnesium Glycinate (Magnesium) | Needed by every organ in the body, helps to build bones and teeth. Helps with regulation of other nutrients such as copper, calcium, potassium, sodium, zinc and vitamin D. Essential for cellular health and contributes to the production of energy. Critical component of over 300 biochemical functions in the body. Blood sugar control. | Infants<br>0-6 months: 100 mg/day<br>7-12 months: 275 mg/day<br>Children<br>1-3 yrs: 420 mg/day<br>4-8 yrs: 500 mg/day<br>9-18 yrs: 1250 mg/day<br>Adults:<br>19-50: 700 mg/day<br>Pregnant or breastfeeding: 700 mg/day | Spinach and dark green leafy vegetables; legumes (chickpeas, lentils, mung beans & black beans); whole grains; almonds, cashews, pumpkin seeds, avos, bananas, raw dark chocolate                                                                                                                                                                                                                                                     | Sleep disorders, restless leg syndrome, irritability, anxiety and agitation, abnormal heart rhythms, low blood pressure, muscle cramping, hyperventilation, confusion, seizures                                                                                                                                   |                                                                                                                                                        | Always give in 1:2 ratio with calcium and add collagen, Vit D3 and K2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Insomnia, muscle cramps, muscle pain, Fibromyalgia and other autoimmune diseases, anxiety, high blood pressure, fatigue, migraines, Osteoporosis, constipation<br>Diabetes, insulin resistance |

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| Vitamin Name | Primary Functions                                                                                                                                                                                                                                                                     | Recommended Intake                                                                                                                                                                                                                                                                                                                                                                                                              | Reliable Food Sources                                                                                                             | Deficiency Symptoms                                                                                                                                                                                                                  | Toxicity Symptoms                                                                                                                                                                                                                                                                              | Interactions with drugs or other vitamins                                                                                                                                                                                                                | Recommend for                                                                                                                                                                                  |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phosphorus   | Builds strong bones and teeth together with calcium and magnesium. Involved in the production of energy for your body's cells. Helps with filtering waste and toxins from kidneys. Plays a key role in the production of many enzymes and hormones, and balances the pH of your body. | <p>Infants:</p> <p>0-12 mths: 400-700 mg/day</p> <p>Children:</p> <p>1-8 yrs: 3000-3800 mg/day</p> <p>9-18 yrs: 4500-4700 mg/day</p> <p>Adults:</p> <p>19+ yrs: 4700 mg/day</p> <p>Pregnant or breastfeeding: 5100 mg/day</p> <p>Children:</p> <p>1-3 yrs = 20 mcg/day</p> <p>4-8 yrs = 30 mcg/day</p> <p>9-13 yrs = 40 mcg/day</p> <p>14 + yrs = 55 mcg/day</p> <p>Pregnant = 60 mcg/day</p> <p>Breastfeeding = 70 mcg/day</p> | Sunflower seeds, white beans, mung beans, tuna, turkey, grass fed beef, almonds, brown rice, potatoes, broccoli, eggs             | Fragile and painful bones, stiff joints; growth and development issues; general weakness, muscle aches and soreness, numbness and general or chronic fatigue syndrome; loss of appetite, anxiety, irritability, irregular breathing. | Diarrhea; hardening of organs and soft tissues; interfering with the balance of iron, calcium, magnesium and zinc which may cause heart complications due to an imbalance in essential minerals that regulate blood pressure, circulation and kidney function. Limiting vitamin D absorability | Works with calcium to build bones, but very high levels of calcium blocks absorption of phosphorus. Interactions with antacids, blood pressure medicines, corticosteroids and insulin                                                                    | Strong teeth and bones, fatigue                                                                                                                                                                |
| Potassium    | Ensures proper functioning of body's tissues and cells. Plays an important role in functioning of the heart. Assists with skeletal and smooth muscle contraction, important for proper digestive function                                                                             | <p>Infants:</p> <p>0-6 mths: 2 mg/day</p> <p>7-12 mths: 3 mg/day</p> <p>Children:</p> <p>1-3 yrs = 3 mg/day</p> <p>4-8 yrs = 5 mg/day</p> <p>9-13 yrs = 8 mg/day</p> <p>Adults:</p> <p>Males 14 yrs+ = 11 mg/day</p> <p>Females 14 -18 yrs = 9 mg/day</p> <p>Females 19 yrs + = 8 mg/day</p>                                                                                                                                    | Citrus, bananas, melons, avos, toamatoes, potatoes, fish, chicken , red meat                                                      | Leg and arm muscle cramps, fatigue, nasea, vomiting, constipation, bloating, abdominal cramping, excessive urination/thirst, low blood pressure, fainting, psychosis, depression, delerium.                                          | Too much potassium may be caused by uncontrolled diabetes, burns or severe injuries, kidney disease, Addisons disease. Toxicity symptoms include heart palpitations, shortness of breath, chest pain, nausea, or vomiting                                                                      | Too much sodium causes low potassium levels; diuretics and laxatives causes low potassium levels; blood pressure lowering medicines might cause too much potassium in the blood; contra-indicated for people with kidney problems or on cancer treatment | Healthy heart rhythm; high blood pressure; risk of stroke; endurance athletes for cramps & dehydration; vomiting and diarrhea; PMS; proper food processing and growth, osteoporosis protection |
| Selenium     | An antioxidant, helps prevent damage from free radicals. Helps with correct functioning of the thyroid and normal immune system functioning                                                                                                                                           | <p>Children:</p> <p>1-3 yrs = 20 mcg/day</p> <p>4-8 yrs = 30 mcg/day</p> <p>9-13 yrs = 40 mcg/day</p> <p>Adults:</p> <p>14 + yrs = 55 mcg/day</p> <p>Pregnant = 60 mcg/day</p> <p>Breastfeeding = 70 mcg/day</p>                                                                                                                                                                                                                | Wheat germ, brewer's yeast, liver, tuna, mackerel, herring, shellfish, butter, whole grains, garlic, brazil nuts, sunflower seeds | Associated with infertility in men, can aggravate iodine deficiency                                                                                                                                                                  | Doses over 400 mcg can cause bad breath, fever, nausea and potential liver & kidney complications, heart attack and respiratory depression.                                                                                                                                                    | Selenium has interactions with antacids, chemotherapy drugs, corticosteroids, niacin, cholesterol-lowering statin drugs and birth control pills.                                                                                                         | Healthy thyroid function; liver, prostate, colorectal and lung cancers; HIV & other viral infections; for fertility, asthma symptoms and longevity.                                            |

## MINERALS

| Vitamin Name                              | Primary Functions                                                                                                                    | Recommended Intake                                                                                                                                                                                                                                                                                                                                                                           | Reliable Food Sources                                                                                                                                                                                                                                            | Deficiency Symptoms                                                                                                                                                                                                                                                        | Toxicity Symptoms                                                                                                                           | Interactions with drugs or other vitamins                                                                                                                                                                               | Recommend for                                                                                                                                                                                                       |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sulphur<br>(methylsulfonylmethane or MSM) | Improves immune function, lowers chronic inflammation, restores healthy connective tissue, lowers muscle pain & spasms               | There is no recommended dietary allowance for sulphur. Dosage depends on what you're using MSM for and current state of health. Usually, 500 mg of MSM, taken two to three times daily, is the starting dose recommended for treating osteoarthritis. Other reports show that up to three to six grams of MSM daily (divided into three doses normally) seems to be safe and well-tolerated. | Dietary sulphur is provided by the sulphur amino acids methionine and cysteine found in protein rich foods such as eggs, meat, dairy, fish, nuts, seeds, whole grains and legumes. Also found in onions, garlic, ginger, horseradish and cruciferous vegetables. | No proven symptoms of sulphur deficiency described in medical literature yet                                                                                                                                                                                               | Mild side effects for some people, including indigestion, upset stomach, diarrhoea, and skin and eye irritation                             | Not suitable for people taking blood-thinning medications                                                                                                                                                               | Osteoarthritis, rheumatoid arthritis & joint pain, osteoporosis, digestive problems and leaky gut, skin conditions, for allergies and asthma, recovery after exercise, auto-immune disorders                        |
| Zinc                                      | Helps with hormone production; growth and repair processes; improves immunity and facilitates digestion; act as an anti-inflammatory | <p>Infants:</p> <p>0-6 mnths: 2 mg/day</p> <p>7-12 mnths: 3 mg/day</p> <p>Children:</p> <p>1-3 yrs = 3 mg/day</p> <p>4-8 yrs = 5 mg/day</p> <p>9-13 yrs = 8 mg/day</p> <p>Adults:</p> <p>Males 14 yrs+ = 11 mg/day</p> <p>Females 14-18 yrs = 9 mg/day</p> <p>Females 19 yrs + = 8 mg/day</p>                                                                                                | Oysters, shellfish, lamb, grassfed beef, chickpeas, cashews, pumpkin seeds sprouted nuts and seeds, chia seeds, flax seeds, sesame seeds, tahini, almonds, peas, kefir                                                                                           | Leaky gut syndrome, persistent diarrhea, compromised immunity, thinning hair or hair loss, inflammatory skin issues, menopausal or PMS symptoms, infertility. Changes in appetite, including food cravings for salty or sweet foods. Changes in ability to taste and smell | Doses over 400 mcg can cause bad breath, fever, nausea and potential liver & kidney complications, heart attack and respiratory depression. | Excessive intake of zinc can suppress copper and iron absorption. These minerals compete for the same absorption sites as they are all positively charged. Zinc and iron should be taken at different times of the day. | Acne, eczema, infertility, recurrent viral infections, wound healing, vegans & vegetarians, persons with digestive issues, alcoholics, women taking the birth control pill or hormone replacement therapy, diabetes |

## 3. ANTIOXIDANTS

### WHAT ARE ANTIOXIDANTS?

Antioxidants are substances that prevent free radicals from attacking and damaging cells. Free radicals are highly unstable molecules that are naturally formed as byproducts of cellular reactions when you breathe, when your body converts food into energy, when you are ill, stress or take part in strenuous exercise. The liver produces and uses free radicals for detoxification, while white blood cells send free radicals to destroy bacteria, viruses and damaged cells. Your body can also be exposed to free radicals from a variety of environmental sources, such as cigarette smoke, air pollution and sunlight.

Why are free radicals thought to be dangerous then? Free radicals are unstable molecules, meaning they're always on the lookout for chemical components that other cells have but that they themselves are missing. Electrons are the negatively charged parts of atoms and exist in pairs, and free radicals are missing an electron. Free radicals "react" with other molecules they come into contact with, robbing one of their electrons to become stable. This process causes the robbed molecule to become unstable and unable to function normally and turns it into an electron-seeking unstable free radical itself, leading to a chain reaction in the body and the proliferation of free radicals.

Free radicals thus cause "oxidative stress," a process that triggers cell damage. Oxidative stress is thought to play a role in a variety of diseases including cancer, cardiovascular diseases, diabetes, Alzheimer's disease, Parkinson's disease, and eye diseases such as cataracts and age-related macular degeneration.

The protective antioxidants mainly include **vitamins A, C, and E, as well as zinc,, selenium, CoQ10, and phytonutrients.**

### WHAT ARE PHYTONUTRIENTS?

Plant foods contain more than 25 000 natural chemicals that are called phytonutrients or phyto-chemicals. "Phyto" refers to the Greek word for plant. They're concentrated in the skin of many fruits and vegetables, and are responsible for their colour, scent and flavour.

They are basically a plant's immune system, protecting them from environmental threats such as ultraviolet radiation, bacteria, fungi, parasites and insects. Since we are exposed to radiation and various environmental elements just like plants, we need phytonutrients to protect us.

Phytonutrients aren't essential for keeping us alive, but when we eat or drink phytonutrients, they may help prevent disease and keep our bodies working properly.

They are used mostly as plant based antioxidants helping to optimise immune function and reduce the risk of infection and can be anti-microbial, anti-inflammatory, provide protection from degenerative conditions, for allergic reactions, to promote healing, improve circulation and improve regeneration of the skin. They are often used to slow down the signs of ageing.

#### **Flavonoids Family**

- Anthocyanides - blue/red bilberry, blueberries and goji berries
- Catechins in cocoa and green tea
- Bioflavanoids - rutin and hesperidin
- Oligomeric Proanthocyanidines (OPC) - pycnogenol, grape seed extract and hawthorne berries
- Quercetin in onions

## Carotenoid Family

- Beta Carotene - yellow, orange and red fruit and vegetables such as pumpkin, carrot, sweet red peppers, the body converts them to Vitamin A
- Lycopene - in pink fruits such as tomatoes, watermelon and grapefruit
- Lutein & zeaxanthin - goji berries, marigold, dark leafy green vegetables, yellow and orange fruit and veg
- Astaxanthin – pink seafood such as shrimp and salmon

## Others

- Chlorophyll - blue green algae, chlorella, spirulina, barley and wheat grass
- Polyphenol - green tea and olive oil
- Resveratrol in red grapes and berries
- Sulforaphane in cruciferous vegetables such as cabbage, broccoli and Brussel sprouts.

## 4. ESSENTIAL FATTY ACIDS

Essential fatty acids cannot be made by the body so we must take them in from the diet or from supplements. There are two forms of Essential Fatty Acids: Omega 3 and Omega 6. Generally Omega 3 is more important as it has better anti-inflammatory properties. Our diets normally have adequate amounts of Omega 6 found in many of the seed oils. Too much omega 6 in the diet can be a cause of inflammation.

### OMEGA 3

- When omega 3 is sourced from plants such as flax-, hemp-, chia- and pumpkin seeds or walnuts, the body still needs to convert it into the two active ingredients Eicosapentaenoic Acid (EPA) and Docosahexaenoic Acid (DHA).
- When omega 3 is taken in the form of oily fish or fish oil supplement it is already in the active forms of EPA and DHA and so does not need to be converted.
- DHA can be found as a vegan algae supplement
- Omega 3 oils as liquids are very sensitive to damage and in liquid form must be kept refrigerated and never heated.
- The amount of fish oil per capsule eg 1000mg is not as important as the amounts of the active ingredients EPA & DHA

#### Omega 3 is needed for:

- Enhanced immunity
- Joint health
- Hormonal health and balance
- Skin, eye and hair health
- Inflammation and pain
- Brain development and function, learning and memory, mood
- Cardiovascular health eg blood pressure, reduce blood clots, lowers cholesterol



### OMEGA 6

- Found in nut and seed oils particularly flaxseed oil - these oils need to be converted by the body into the active ingredient Gamma-Linolenic Acid (GLA)
- When taken in supplement form as borage or evening primrose oil it is already in the active GLA form

#### GLA can be anti-inflammatory and is good for:

- PMS
- Heavy periods
- Breast Health
- Hormonal Health
- Eczema



### MCT OIL - MEDIUM CHAIN TRIGLYCERIDES

This oil is found in coconut oil but is also now available as pure MCT oils, it is used to make bullet proof coffee.

#### Useful for:

- Improving athletic performance by boosting energy and endurance
- Weight control - helps switch body to ketosis using fat as primary energy source rather than carbohydrates
- When poor absorption of fats is a problem (no gallbladder) MCT is the easiest fat for the body to absorb and utilise
- Recommended for cognitive decline



## 5. AMINO ACIDS

Amino acids are the building blocks of proteins. All proteins when eaten get broken down into amino acids. The body then uses these amino acids for the repair and growth of body tissue.

### They are needed to:

- Build the structure of bones, muscles, ligaments, tendons and skin
- Synthesize hormones, enzymes and neurotransmitters
- Replicate cells, including immune cells
- Enhance fat loss and preserve lean body mass
- Improve athletic performance
- Boost mood and promote better sleep
- For repair and healing the body

With age or poor stomach acid, proteins may not be broken down and therefore cannot be adequately absorbed and used by the body. This may lead to amino acid deficiencies.

## ESSENTIAL AMINO ACIDS

These cannot be made by the body and must come from the diet:

- **Histidine:** Used to produce histamine, a neurotransmitter that is vital to immune response, digestion, sexual function and sleep-wake cycles. It is critical for maintaining the myelin sheath, a protective barrier that surrounds nerve cells.
- **Isoleucine:** One of three branched-chain amino acids, meaning it has a chain branching off to one side of its molecular structure. It is involved in muscle metabolism and is heavily concentrated in muscle tissue. It is important for immune function, haemoglobin production and energy regulation, as well as hormone excretion.
- **Leucine:** A branched-chain amino acid that is critical for protein synthesis and muscle repair. It also helps regulate blood sugar levels, stimulates wound healing and produces growth hormones.
- **Lysine:** Plays a role in protein synthesis, hormone and enzyme production and the absorption of calcium. It is important for energy production, immune function and the production of collagen and elastin. A potent anti-viral.
- **Methionine:** Plays an important role in metabolism and detoxification. It is also necessary for tissue growth and the absorption of zinc and selenium.
- **Phenylalanine:** A precursor for the neuro-transmitters tyrosine, dopamine, epinephrine and norepinephrine. It is integral for the production of other amino acids.
- **Threonine:** A principal part of structural proteins such as collagen and elastin, which are important components of the skin and connective tissue. It also plays a role in fat metabolism and immune function.
- **Tryptophan:** Needed to maintain proper nitrogen balance and is a precursor to 5 HTP, then serotonin, a neurotransmitter that regulates appetite, sleep and mood.
- **Valine:** One of three branched-chain amino acids, it helps stimulate muscle growth and regeneration and is involved in energy production.

Animal products like beef, fish, dairy, and eggs contain enough of every one of these essential amino acids. Thus, they're considered complete proteins.

Plant sources of protein are usually too low in or missing one or more of these essential amino acids. They're considered incomplete protein sources. It is still possible to get enough of each essential amino acid throughout the day by eating a varied diet and combining complementary plant proteins. Good sources of plant proteins are quinoa, amaranth, buckwheat, spirulina, hemp seeds, chia seeds, fortified nutritional yeast, as well as the combination of grains and legumes.

## OTHER IMPORTANT AMINO ACIDS

**L-Arginine** - necessary for the production of nitric oxide which assist in managing blood flow. Helps with blood pressure management, athletic performance and erectile dysfunction

**L-Carnitine** - made from the amino acids lysine and methionine. Helps transport fatty acids into the mitochondria, where they can be burned for energy. Good for energy, athletic performance, heart health.

**Acetyl L-Carnitine** - to enhance brain function, for cognitive decline and ageing

**N-Acetyl Cysteine (NAC)** - along with glutamine and glycine, NAC is needed to make and replenish glutathione. Protects body against environmental toxins and protects liver and kidneys from damage. In chronic respiratory conditions as an expectorant as it thins mucous. For male and female fertility. For neurological disorders such as Alzheimers disease and Parkinson's disease, withdrawal symptoms and addictive behaviour. May thin the blood.

**L-Glutamine** - important role in intestinal health, is an energy source for intestinal and immune cells, repairs gut lining and supports immune health

**L-Glutathione** - comprised of three amino acids: glutamine, glycine, and cysteine. The master antioxidant that reduces oxidative stress and battles inflammation. Protects the liver, and used for auto-immune disease, cognitive decline, autism and cancer.

**Glycine** - one of three amino acids that your body uses to make glutathione, it is also one of three amino acids that your body uses to make a compound called creatine, and the main amino acid in collagen. Used for muscle growth, joint and cartilage repair, digestion. Found in bone broth. Magnesium is often bound (chelated) to glycine for improved effectiveness and absorption

**L-Theanine** - involved in the release of neurotransmitters, including GABA. It has a calming, sedative effect on the body and mind without drowsiness — which is why it's often used to reduce anxiety, hyperactivity and sleep-related problems.

**Taurine** - may improve eyesight and hearing, for anxiety and nervous tension, may help reduce risk of cardiovascular disease, may improve blood sugar control.

**5-Hydroxytryptophan (5-HTP)** – used to produce serotonin to assist with depression, anxiety, sleep disorders and weight gain

**L-Tyrosine** - mental alertness, appetite control and depression, makes dopamine (needed for drive, motivation & resistance to stress)



## 6. PROBIOTICS & DIGESTIVE ENZYMES

### PROBIOTICS

Probiotics are good bacteria essential for a healthy digestive system and for strengthening immunity. They help the body manufacture some vitamins such as vitamins K and B12 and increase the bioavailability of minerals.

Probiotics may also help boost mood and cognitive function. Research shows that the gut and brain are connected physically and chemically, a partnership called the gut-brain axis. Changes in the gut can directly affect the brain. The brain and gut communicate through the vagus nerve as well as gut microbes, which produce molecules that carry information from the enteric nervous system in the gut to the brain.

The term "dysbiosis" refers to when the intestines and gut bacteria are in a diseased state. This may be due to the presence and overgrowth of unfriendly bacteria, which may lead to chronic inflammation. Researchers have identified gut dysbiosis in people with obesity, high cholesterol levels, heart disease and type 2 diabetes.

Probiotics may also benefit those with skin conditions such as eczema and dermatitis. Consider replenishing the microbiome on the skin with skin care products that contain probiotics, and by minimising the use of anti-bacterial products.

#### Recommend probiotics for:

- **Digestive health** - chronic indigestion, abdominal pain or bloating, food intolerance or allergies, diarrhoea and travellers diarrhoea, constipation, IBS, restoring the intestines after taking antibiotics or after a bout of food poisoning, to inhibit overgrowth of pathogenic bacteria and yeast such as candida
- **Immune health** - children who have to attend creches, persons who constantly have a cold, the flu or an infection of some sort.
- **Cognitive health** - low mood, anxiety, depression, ADHD, autism, dementia, Alzheimer's, Parkinson's
- **Metabolic health** - overweight, obesity, high cholesterol, heart disease, insulin resistance and type 2 diabetes
- **Skin health** - eczema and psoriasis

Probiotics are found in fermented foods and drinks such as yoghurt, kefir, sauerkraut, kimchi, kombucha, apple cider vinegar and miso.

When using a probiotic supplement always look for probiotics that have identifiable strains that have been clinically proven to be safe and effective for the intended application and have a guaranteed dosage at date of expiry.

Prebiotics are soluble fibers that will help with the growth and proliferation of good gut microbes, they are often added to a probiotic supplement.



## How to take probiotics:

- Start with a small dose. Every person's microbiome is different. When starting to add these foods or supplements to the diet, start with small amounts and steadily build up to the recommended dosage. Experiencing gas, bloating, and increased bowel activity isn't uncommon when adding probiotics to the diet for the first time.
- Avoid taking probiotics with chlorinated water. Chlorine is added to municipal water systems to kill bacteria, and this could very well impact the effectiveness of your probiotic supplement.
- Probiotics are usually best consumed on an empty stomach, when stomach acid levels are lower.
- Make sure to provide a nurturing environment for probiotic organisms by consuming adequate amounts of prebiotic fibre, or taking a supplement that contains a synergistic blend of pre- and probiotics in the same product.
- Never take antibiotics and probiotics at the same time of day, as the antibiotics may kill the beneficial bacteria in the probiotics

## DIGESTIVE ENZYMES

Digestive enzymes help the digestive system to break down proteins, fats and carbohydrates so that they can be absorbed and used by the body. Beneficial bacteria in the large intestine also produce enzymes that aid the digestive system in breaking down complex plant fibre that may be difficult to digest.

Proteases break down proteins, carbohydrases break down carbohydrates and lipases break down fats.

**Protease enzymes** are secreted by the stomach, pancreas and small intestine. An example is pepsin, which is secreted in the stomach and works best when the stomach produces enough hydrochloric acid. Proteins are long chains of amino acids, and protease enzymes break them into peptides and eventually into individual amino acids, which are small and easily absorbed in the small intestine. Some proteolytic enzymes such as bromelain and papain may help to reduce inflammation, assist with healing and recovery from wounds and reduce sore muscles and joints. Use them in between meals to have the best effect.

**Carbohydrase enzymes** are secreted by the mouth, pancreas and small intestine. An example is amylase which is secreted by the mouth and found in saliva, it starts to work as soon as we begin to chew our food. Amylase digests long, complex starch molecules into smaller, simpler maltose molecules which is further broken down into glucose by the enzyme maltase. Other dietary sugars such as sucrose and lactose are broken down further by the carbohydrase enzymes sucrase and lactase.

**Lipase enzymes** are secreted by the pancreas and the walls of the small intestine. They digest complex fat (or lipid) molecules into simple, soluble fatty acid and glycerol molecules.

As we age and because we stress, digestive enzyme and hydrochloric acid secretion may be reduced leading to lower nutrient absorption. Supplementation with a complex of different digestive enzymes should be recommended. Some products have added betaine hydrochloride to increase stomach acid, as well as specific enzymes to help break down plant fibre and the lactose in dairy. They may also contain bitter and aromatic digestive herbs such as fennel, liquorice and peppermint which are known to stimulate appetite and the body's ability to produce its own digestive enzymes, as well as aid the flow of bile and support the liver in its detoxification work. It is the bitter taste in the mouth that stimulates the body into digestive action - recommend that bitter coffee, chocolate, citrus peel pith and bitter leaves such as dandelion, arugula (rocket), radicchio and fennel are included in the diet.

Some products contain enzymes from animal sources, eg. ox bile - for vegan and vegetarian customers this will be unacceptable. Vegetarian sources of enzymes include pineapple (protease), papaya (protease), mango (amylase), banana (carbohydrase), avocado (lipase), kiwi (protease), ginger (protease), apricot (carbohydrase) and raw honey (variety of enzymes).

Fermented foods like kimchi, sauerkraut, tempeh, and miso typically include all the enzymes required to break down proteins, fats and carbohydrates.



## 7. NATURAL FOOD SUPPLEMENTS

### ALOE VERA

- Can be used topically and as a drink
- Skin healing and soothing for bites, stings, sunburn, eczema, acne and rashes
- Gut healing for stomach ulcers, constipation and haemorrhoids
- Gentle colon cleansing and detoxification
- Restore skin elasticity



### CHARCOAL

- Activated charcoal usually from coconut shells
- Used to trap toxins and chemicals in the digestive system allowing them to be flushed out so the body does not reabsorb them
- Treatment of food poisoning, stomach bug, drug overdose or excess alcohol consumption
- Useful for treating diarrhoea
- Not for long term use as it might adsorb medications and important nutrients

### CHLORELLA

- Rich in vitamins, minerals and amino acids, especially good source of plant-based vitamin B12
- Important to use for detoxification especially of heavy metals (use together with Zeolite to trap the heavy metals and escort them out of the body)

### COLLAGEN

Collagen is one of the most abundant proteins found in our bodies. In supplement form it is derived from the connective tissues of animals such as chicken, pork, beef and fish.

Foods that contain gelatin, such as bone broth, also provide collagen. Gelatin is a protein substance derived from collagen after it has been cooked.

Collagen in supplements has already been broken down, or hydrolyzed, to be easily absorbed.

People with allergies to fish, shellfish and eggs should avoid collagen supplements made with these ingredients to prevent allergic reactions.

Recommend vitamin C and MSM to assist with collagen synthesis in the body, or for vegans and vegetarians.

- Improves health of skin, nails and hair
- Increases muscle mass, prevents bone loss and relieves joint pain
- Helps to heal the gut lining



## COQ10

- Is an antioxidant but is also critical for cellular energy production
- Important for cardiovascular health
- Helps metabolise fats and proteins
- Anti-ageing
- Helps improve gum health
- Important to use if on Statins or Red Yeast Rice products



## COLLOSTRUM

- The first breast milk is high in colostrum, usually the source is bovine (cow)
- Contains high levels of protein and growth factors
- Helps to support the immune system
- Helps accelerate healing
- Increase vitality and stamina
- Helps to heal leaky gut



## GARLIC

- Helps to lower blood pressure
- Reduce risk of blood clots, heart attacks
- Lowers cholesterol
- Immune system support
- Natural anti-microbial - can be used for viral, bacterial and fungal infections



## GRASSES (WHEAT & BARLEY)

- Rich in vitamins, minerals, chlorophyll and enzymes
- Can be used as a plant-based food supplement
- Helps to heal the digestive system and reduces inflammation
- Alkalising, cleansing, detoxifying and healing

## KELP

- Can be used as a dietary supplement especially for mineral deficiencies
- Rich source of iodine, helps with low thyroid function
- Useful for weight loss, helps to increase metabolism



## MSM - METHYLSULFONYLMETHANE

- Rich in sulphur containing compounds
- Nourishes hair, skin and nails
- Helps heal injuries
- Detoxification - start off slowly as headaches may be experienced
- Helps to maintain healthy joints and for pain relief
- Important for respiratory health, improves breathing and helps for asthma
- Involved in proper insulin function and glucose metabolism



## PEA PROTEIN

- Good source of plant based amino acids
- Important for vegans
- Improves muscle mass as it contains BCAA
- Good for cardiovascular support



## PHOSPHATIDYL CHOLINE (PC) AND SERINE (PS)

- PC is a component of lecithin, helps break down fats
- Prevents hardening of the arteries
- Helps support the liver
- PS is important for cell membranes especially nerve cells
- Helps to improve memory loss, concentration and depression



## SPIRULINA

- A fresh-water blue-green algae
- Rich in nutrients including protein, essential omegas, vitamins and minerals
- Useful dietary food supplement for vegans and vegetarians
- Helps support the immune system
- Helps to cleanse and heal the body
- Helps to curb appetite and assists with blood sugar balance
- Helps increase energy and vitality



## WHEY PROTEIN

- It is the liquid left over from cheese making, rich in protein, free from fat and lactose
- Important for helping to build lean muscle
- Useful for athletes
- Help to protect against muscle wasting eg post chemotherapy
- Helps to enhance immune function



# QUESTIONS:

1. Name 3 reasons why we would need supplementation even when we consume a healthy and balanced diet
2. How does a nutraceutical differ from a supplement?
3. What does the "% of Daily Value" on a supplement label mean?
4. Please complete: The abbreviation "mg" stands for \_\_\_\_\_, or one thousandths of a gram, and "mcg" stand for \_\_\_\_\_, or one thousandths of a milligram. The measurement symbol "μ" means \_\_\_\_ in Greek and is the same as the measurement microgram. IU stands for \_\_\_\_ and is used as a measurement of \_\_\_\_ vitamins.
5. True or False: You may recommend to a customer to exceed the dosages as stated on the supplement label.
6. Name the fat-soluble vitamins and describe how they are absorbed into the body
7. True or False: Fat soluble vitamins are safe to take in huge doses. Why?
8. Name the water-soluble vitamins and describe how they are absorbed into the body
9. Name 3 reasons why you would recommend Vitamin D3 to a customer
10. Name 3 reasons why you would recommend Vitamin C to a customer
11. Name 3 conditions you would recommend a B-vitamin complex for
12. Which B-vitamin would you recommend to vegans and vegetarians? Why?
13. What are minerals? Name 3 minerals that is taken at dosages of more than 100mg/day
14. What are trace minerals? Name 3 trace minerals
15. In which form are minerals easiest absorbed by the body?
16. What are chelated minerals?
17. What are slow release vitamins and minerals?
18. In which ratio would you recommend that Calcium and Magnesium be taken?
19. Which two fat soluble vitamins are recommended to be taken with Calcium and Magnesium? Why?
20. Name 5 reasons for taking Selenium
21. Name 5 reasons for taking Zinc
22. What are Anti-oxidants?
23. What are Free radicals and how are they generated?
24. What are Phyto-nutrients? Why are they important?
25. What are Essential Fatty Acids? Name the two forms
26. Name 5 reasons why everybody should be taking an Omega 3 supplement.
27. Name 3 sources of Omega 3 where these essential fatty acids still need to be converted to EPA and DHA?
28. From which source can omega 3 be taken already in the active EPA and DHA form?
29. Name two sources that provide omega 6 in the active GLA form
30. Name 3 conditions that Omega 6 may be recommended for
31. When will you recommend MCT oil to a customer?
32. What does MCT stand for?
33. What are amino acids?

## QUESTIONS:

34. What are essential amino acids?

35. Name the main benefits for the following amino acids:

## L-Arginine -

BCAA -

## Acetyl L-Carnitine -

## L-Carnitine -

## N-Acetyl Cysteine (NAC) -

## L-Glutamine -

## L-Glutathione -

Glycine -

## L-Lysine –

## Taurine -

## L-Tryptophan -

## L-Tyrosine –

## L-Theanine -

### 36. What are probiotics?

### 37. Why are probiotics important to our health?

38. Name 5 foods that contain probiotics

39. What is the function of digestive enzymes in the body?

40. What is the role of hydrochloric acid in the body?

41. True or False: Activated Charcoal can be taken for long periods of time. Why?

#### 42. Why is Co-Enzyme Q10 important?

43. What is Collagen? What can it be recommended for?

44. Name 6 medicinal mushrooms, and explain how they support the body

45. What is MSM, which foods contain sulphur naturally and for which conditions can we recommend it for?

[illegible]

# 6

## MODULE SIX

### HERBAL MEDICINE

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*"When diet is wrong, medicine is of no use. When diet is correct, medicine is of no need"*

- Ancient Ayurvedic proverb

The use of herbs as a source of healing is inherent in all cultures and throughout time. It is our oldest form of healing. Herbal remedies enhance other natural therapies to treat a wide range of conditions, from the common cold to more serious chronic conditions.

**Phyto-therapy** is a form of natural medicine - it is a science based medical system using plant-derived medication to treat and prevent disease.

## AT WELLNESS WAREHOUSE WE SELL HERBAL REMEDIES FROM FOUR TRADITIONS OF HERBAL MEDICINE:

- **Traditional Chinese Medicine** is based on more than 2,500 years of medical practice that includes the use of herbal medicine, acupuncture, cupping therapy, massage therapy and spiritual movement practices such as qigong and tai chi. Key principles are to treat the whole body as an integrated system, the knowledge that the body has a natural self-healing ability, understanding that we are connected to nature and we are influenced by it, and that prevention is the best cure.

- **Ayurvedic medicine** originated in India more than 3000 years ago. In Sanskrit, Ayurveda means "The science of life". Ayurveda places great emphasis on prevention and encourages the maintenance of health through close attention to balance in one's life, right thinking, diet, lifestyle and the use of herbs. Knowledge of Ayurveda enables one to understand how to create this balance of body, mind and consciousness according to one's own individual constitution and how to make lifestyle changes to bring about and maintain this balance.

- **The Unani Tibb tradition** is a system of medicine that originated in Africa, was embraced by Greeks like Hippocrates and refined over many centuries by Arab physicians like Ibn Sina. It recognises each person's uniqueness, known as their temperament, by describing a person's physical characteristics and their psychological and emotional attributes. It assesses personality strengths and weaknesses, and includes the predisposition for particular disorders. Herbs, cupping, meditation, massage, aromatherapy and colour therapy are used to achieve balance.

- **Western herbalism** draws from the herbal traditions of Europe and the Americas, and emphasizes the study and use of European and Native American herbs in the treatment and prevention of illness. This system acknowledges the affinity of herbs to certain patterns of disease (heat/wind/dry/damp), understands that the body works as a whole and that herbs have affinity to certain organs or systems, and lastly uses herbs to assist the body in its patterns of self-regulation.

Herbal Medicine uses both herbs (leaves, flowers and roots) and spices (seeds, berries, fruits, bark and root).



## 1. WAYS TO USE HERBS

**Teas:** An infusion is the simplest form of herbal preparation. If you know how to make a cup of tea, you know how to make an infusion! Both fresh and dried herbs may be used to make teas.

**Tinctures:** Tinctures are concentrated forms of medicinal herbs, made by extracting the active constituents with a mixture of water and alcohol. Herbs are absorbed more quickly into circulation, and therefore begin to take effect faster when taken in tincture form. They can also be preserved without alcohol.

**Capsules & tablets:** Dried or powdered herbs can be taken in capsules or tablets. Capsules have the advantage of being easy and convenient, especially for bitter-tasting herbs. The disadvantage of capsules is that the herbs inside them are harder to assimilate into the body than preparations made with water or alcohol, such as tinctures.

**Ointments (topical):** Herbal creams and ointments help for localised skin or muscle complaints, such as eczema, wound healing, fungal infections, rashes, bruises and sore muscles. Herbs are gently heated in oils, fats or cream bases to extract the active compounds. Examples are Calendula and Arnica creams and oils, as well as menopausal creams containing natural progesterone from Wild yams.

**Essential Oils:** Very concentrated herb oils

## WHOLE HERBS

A whole herb is the complete plant dried and encapsulated or prepared as a tincture. Whole herbs contain all of the complex constituents of the plant and have been used in this way for thousands of years by many cultures. The medicinal properties of herbs were found through intuitive observation of their use and the information has been passed down through successive generations of healers. Herbs used in their whole form have the ability to address multiple health issues simultaneously in ways that we have not even thought of or studied yet.

There are certain things that may affect the components of whole herbs - the soil it's grown in, the types of pest-control used, the time of year the herb is harvested, the weather patterns and the air quality are only some of the factors that influence individual plants. It is important to consider the part of the plant that is extracted – root, stem, leaves, fruit – and the processing method employed. Every farm and every manufacturer is different. Even within the same farm or source, generations of a plant may evolve. In other words, whole herbs do not guarantee uniformity and, therefore, may have varying levels of potency and health benefits. Always make sure to choose reputable suppliers that source herbs that are grown organically and process plants in pharmaceutical grade facilities.

## STANDARDISED HERBS

A standardised herb is a herb extract that has one or more components present to ensure consistency in amounts, quality and activity. This practice has developed out of the drug model of herbal medicine, in which modern scientists have attempted to identify the components of a plant that have definite pharmacological activity in the body. Unfortunately, while scientists can isolate many constituents from a herb and discover how particular chemicals may act in the body, they inadvertently remove or overlook components that may contribute to the activity of the whole herb. Consequently, standardisation may concentrate one constituent at the expense of other potentially important ones, while changing the natural balance of the herb's components.

Standardisation, therefore, is based on the idea that isolated compounds are responsible for the action of a herb. Almost no medicinal herbs, however, are known for just a single function. In fact, plants contain a complex blend of phytochemicals, and as naturally concentrated foods have the unique ability to address a multiplicity of problems simultaneously. The full medicinal value of herbs is most likely due to their internal complexity and to the interactions of the different components within the body rather than to one of its specific components.

Furthermore, many of the constituents in herbs are as yet unknown, and internal chemical interactions within and among herbs are even more poorly understood. Therefore, it is doubtful that this type of standardised herbal extract can exhibit the same full spectrum of use as the whole herb.

However, science has proven the efficacy of some of these concentrated extracts regarding specific biological actions, so they can be very useful even if they don't work exactly like the whole herbs. Be careful not to recommend their use over long periods of time or in large doses.

## HERBAL FORMULAS

It is often very beneficial to combine herbs that have the same actions in the body or address the same organ system or ailment, for example an immune blend that contains the synergistic blend of the herbs Echinacea, Ginger, Cayenne, Red Clover and Elder Flower.

Herbs may be combined to great effect with nutrients that enhance the actions of the herb - for example, an immune blend that contains Elderberry, Medicinal Mushrooms, vitamins C and D, Zinc and Selenium.

## 2. CONTRA-INDICATIONS

Although herbs are considered natural, they are very powerful medicines and might interact with or interfere with the effectiveness of chronic medicines. Some herbs, including garlic, ginkgo, ginseng, and St. John's Wort, can have a significant influence on chronic medicines.

Herbal medicines may mimic, decrease, or increase the action of prescribed drugs. This can be especially important for drugs with narrow therapeutic windows and in sensitive patient populations such as older adults, the chronically ill, and those with compromised immune systems.

**Always recommend not taking herbs and chronic medicines together, and to keep monitoring for potential interactions by maintaining routine testing, such as blood coagulation tests given to users of Warfarin. Always recommend that customers should only take herbs with chronic medicines under supervision of their physician. It is important to slowly phase in the use of herbs while systematically lessening the amount of chronic medicines used. Never recommend that chronic medicines be stopped abruptly.**

## THE MOST IMPORTANT INTERACTIONS WITH CHRONIC MEDICINES TO LOOK OUT FOR ARE:

### i) Blood thinning medication

Blood thinners are medications taken orally or intravenously (through a vein) to prevent a blood clot. Blood clots can stop the flow of blood to the heart, lungs, or brain and may cause a heart attack or stroke. Blood thinners are usually prescribed for heart disease and irregular heart rhythms, thrombosis and strokes.

Blood thinners are divided into anti-platelet and anti-coagulant drugs.

Anti-platelet drugs prevent blood cells (called platelets) from clumping together and forming clots. Aspirin and Ecotrin are a good examples.

Anticoagulants prevent blood clots by increasing the amount of time it takes the blood to clot. Examples are Warfarin, Heparin, Coumadin, Xarelto, Pradaxa and Eliquis.

Herbs may interact with blood thinning drugs in different ways. They may have anti-platelet or anti-coagulant activity and some herbs and foods might affect blood thinning by containing high amounts of vitamin K.

Herbs that may interact with blood thinners: bilberry, black seed oil, bromelain, cayenne, chamomile, curcumin, dong quai, feverfew, garlic, ginger, ginkgo biloba, ginseng, goldenseal, green tea, hawthorne, horse chestnut, liquorice, meadowsweet, motherwort, St John's wort.

Essential oils that may interact: sweet birch, garlic, oregano, thyme, tarragon and wintergreen.

Remember that Vitamin E, omega 3 oils, as well as flaxseed oil and evening primrose oil may have blood thinning effects and should not be taken in high doses over long periods of time if blood thinning drugs are taken.

Excessive bleeding may occur in a variety of ways, including: heavy periods, bloody or discolored urine or feces, nosebleeds, bleeding gums and prolonged bleeding from a cut or wound.

Vitamin K1 has the opposite effect of blood thinners and helps the blood to coagulate to enable cuts and wounds to stop bleeding. Vitamin K1 is found in leafy greens (spinach, kale, swiss chard), cruciferous veg (cabbage, broccoli, cauliflower), other green veg (asparagus, kelp, cucumbers) and green superfood powders (spirulina, moringa, barleygrass, wheatgrass, chlorella).

## **ii) Anti-inflammatory medication**

Nonsteroidal anti-inflammatory drugs (NSAIDs) are drugs that help reduce inflammation, which often helps to relieve pain. They are used for headache, backache, muscle aches, inflammation and stiffness caused by arthritis and other inflammatory conditions, menstrual aches and pains, pain after a minor surgery, as well as sprains or other injuries. They are often bought as over-the-counter (OTC) medications without a doctor's prescription, which may lead to overuse and adverse side effects.

Brand names of NSAIDS include: Aspirin, Voltarin, Celebrex, Advil, Brufen and Myprodol.

Anti-inflammatory drugs may thin the blood, so they should not be used together with herbs that have anti-platelet activity (ginkgo biloba, garlic, ginger, bilberry, dong quai, feverfew, ginseng, turmeric, meadowsweet and willow), with those containing coumarin (chamomile, motherwort, horse chestnut, fenugreek and red clover) and with tamarind, enhancing the risk of bleeding.

## **iii) Diabetes medication**

Herbs and spices that might interact with diabetes medication: Aloe vera, berberine, fenugreek, gymnema, curcumin, ginger, garlic, black seed, Korean ginseng, American ginseng, bitter melon, St John's wort, astragalus.

# **USE OF HERBS IN PREGNANCY:**

Little is known scientifically about the safety of most herbs during pregnancy, as most have not been formally evaluated and ethical considerations limit human clinical investigation (testing) during pregnancy.

A phytotherapist, midwife, naturopathic or integrative physician trained in the use of botanicals during pregnancy should be consulted when using herbs medically.

**The safest approach to the use of herbs during pregnancy is to avoid herbs during the first trimester and after this to use herbs that are prescribed by a customer's health care provider.**

**Herbs known either scientifically or historically to be safe during pregnancy:**

### Herbal teas for minor discomforts:

Herbal teas that are known to be safe in moderate amounts (spearmint, German chamomile, lemon balm, nettles, rose hips, ginger) can be considered for use in pregnancy. These herbs help manage nausea, ease digestion and constipation, relieve heartburn, reduce insomnia, and promote gentle relaxation.

### Daily nutritive tonic teas:

A number of herbs are composed of highly nutritious substances such as proteins, carbohydrates, vitamins, and minerals and which can be used safely in pregnancy as daily nutritive tonics.

Traditional herbalists believe that these herbs help prevent anaemia and other mineral and vitamin deficiencies, for example, alfalfa, stinging nettles, milky oats, dandelion root and leaves and red raspberry leaf.

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## 2. HERBS & THEIR USES

Common culinary herbs and herbs proven to be effective for the following herbal actions or concerns:

### **ADAPTOGENIC - for stress**

Ashwagandha, holy basil (tulsi), maca, liquorice, Panax ginseng, reishi, Rhodiola rosea, Schisandra

### **ALLERGIES - environmental allergies**

Goldenseal, green tea, stinging nettle

### **ANTI-BACTERIAL**

Echinacea, garlic, goldenseal, myrrh, olive leaf, oregano oil, thyme

### **ANTI-FUNGAL**

Coconut oil, garlic, grapefruit seed extract, oregano oil, Pau d'Arco

### **ANTI-INFLAMMATORY**

Boswellia, cayenne, chamomile, cinnamon, devil's claw, garlic, ginger, turmeric

### **ANTI-PARASITIC**

Artemisia (wormwood), black walnut, garlic

### **ANTISPASMODIC – muscles or colon**

Chamomile, ginger, peppermint

### **ANTI-VIRAL**

Andrographis, cat's claw, echinacea, elderberry, garlic, ginger, goldenseal, olive leaf, peppermint, sage, stinging nettle, sutherlandia, thyme

### **BLOOD SUGAR**

Aloe vera, bitter melon, cinnamon, fenugreek, green tea, gymnema

### **CALMING**

Chamomile, lemon balm (melissa), passionflower, scelletium, scullcap, valerian

### **CARDIOVASCULAR**

Astragalus, garlic, green tea, hawthorn, hibiscus

### **CIRCULATION**

Cayenne, garlic, ginger, ginkgo biloba, gotu kola, rosemary

### **COGNITIVE/CONCENTRATION**

Bacopa, basil, ginkgo biloba, Rhodiola rosea, rosemary

### **DETOXIFICATION**

Dandelion root and leaf, green tea, green superfood powders, milk thistle, turmeric

### **DIGESTION**

Dandelion leaf, fennel, fenugreek, ginger, liquorice, peppermint, swedish bitters, turmeric

### **DIURETIC - WATER RETENTION**

Astragalus, buchu, dandelion leaf, hawthorne, hibiscus, juniper, parsley

### **ENERGISING**

Ashwagandha, ginkgo biloba, Panax ginseng



### EYE HEALTH

Bilberry, eyebright, ginkgo biloba

### FEMALE HEALTH

Black cohosh, dong quai, chasteberry (Vitex agnus-castus), red clover, sage, saw palmetto

### IMMUNITY

Astragalus, black cumin seed, echinacea, elderberry, medicinal mushrooms, Panax ginseng, sutherlandia, Siberian ginseng

### MALE HEALTH

Ashwagandha, hawthorn, maca, nettle root, saw palmetto

### MOOD ENHANCER

Rhodiola rosea, scutellaria, St. John's wort, saffron

### PAIN RELIEF

Cayenne, devil's claw, feverfew, turmeric, white willow

### RESPIRATORY

Elecampane, eucalyptus, goldenseal, mullein, oregano oil, pelargonium, thyme

### SKIN

Aloe vera, burdock, chamomile, echinacea, gotu kola, horsetail, lavender

### URINARY TRACT - BLADDER

Bilberry, buchu, cranberry, echinacea, goldenseal, uva ursi (bearberry)

### VEIN SUPPORT

Bilberry, butcher's broom, calendula, gotu kola, horse chestnut, witch hazel

### VITAMIN C

Acerola, baobab, camu camu, bilberry, elderberry, hibiscus, rosehip



## 3. ESSENTIAL OILS

Essential oils are produced when the beneficial compounds of the plant are extracted by distillation. This concentrates the benefits of the plant, so only a few drops of essential oils are needed.

### ESSENTIAL OIL APPLICATIONS:

- **Baths and foot baths** - add 5 - 10 drops of essential oil into running water and swish to disperse
- **Inhalations** - add 4 drops of essential oil into a bowl of steaming hot water, place a towel over the head to capture the steam. Close eyes and inhale
- **Humidifier** - add 3 - 9 drops of essential oil into the water of a humidifier
- **Massage** - 20 drops of essential oil in 50ml of base oil
- **Room sprays** - Add 40 - 60 drops of essential oil to water in a spray container shake before use
- **Burners** - add a few drops to the burner with water or coarse salt depending on the burner
- **Compresses** - add 5 drops of essential oil to a bowl of warm water swish cloth in water and wring out, place on affected area

## HOW TO USE ESSENTIAL OILS:

- Essential oils should always be diluted in carrier/base oils before application on the skin
- Essential oils should not be taken internally
- Avoid using around the eyes and other sensitive areas
- Infants (0 - 12 months): Use 1 drop of lavender, chamomile or mandarin diluted in 1 tablespoon of base oil
- Babies (1 - 5 years): Use 2 - 3 drops of safe oils diluted in 1 tablespoon base oil

**NB Essential oils should only be used during pregnancy with the aid of an aromatherapist.**

## RECOMMEND ESSENTIAL OILS FOR:

- ACNE** - chamomile, geranium, lavender, patchouli, vetiver and palmarosa  
**ANXIETY** - basil, bergamot, sandalwood, geranium, neroli, orange and rose  
**BRONCHITIS** - eucalyptus, frankincense, rosemary and pine  
**BRUISES** - cypress, fennel, lavender and tea tree  
**BURNS** - lavender and chamomile  
**CELLULITE** - cypress, juniper, grapefruit and rosemary  
**CIRCULATION** - ginger, vetiver, cypress, rosemary and black pepper  
**COLDS** - lemon, eucalyptus, thyme, pine and tea tree  
**CONCENTRATION** - cedarwood, frankincense, patchouli and rosemary  
**COUGHS** - eucalyptus, thyme, sandalwood and lemon  
**FATIGUE** - geranium, peppermint, basil, palmarosa and clary sage  
**FUNGAL INFECTIONS** - lemongrass, tea tree, lavender and thyme  
**HEADACHE** - lavender, lemongrass, peppermint, melissa and rosemary  
**INSECT BITES** - lavender, melissa and tea tree  
**INSECT REPELLANT** - citronella, lemongrass, lavender, cinnamon, lemon and peppermint  
**INSOMNIA** - chamomile, lavender, melissa, neroli, sandalwood and ylang ylang  
**JOINT PAIN** - frankincense, ginger, rosemary, black pepper, cypress and pine  
**MENOPAUSE** - clary sage, geranium, lavender, neroli, peppermint, rose  
**MUSCLE ACHES** - rosemary, eucalyptus, wintergreen, marjoram, lavender and peppermint  
**PMS** - clary sage, geranium, orange, mandarin, lemon and melissa  
**SINUSITIS** - rosemary, eucalyptus, peppermint, pine and lemon  
**SUNBURN** - lavender, chamomile and peppermint  
**VARICOSE VEINS** - cypress, rosemary, lavender and geranium  
**WARTS** - lavender, lemon, oregano, tea tree



## CARRIER OILS/BASE OILS

- Almond Oil/Sweet Almond Oil
- Apricot Kernel Oil
- Argan Oil
- Avocado Oil
- Baobab Oil
- Coconut Oil
- Grapeseed Oil
- Hemp Oil
- Jojoba Oil
- Olive Oil
- Rosehip Oil
- Sesame Oil
- Shea Butter
- Wheatgerm Oil



## COMMON CONTRA-INDICATIONS

- **Oils not to be used in pregnancy and on babies:** basil, clary sage, hyssop, juniper, marjoram, myrrh, sage, fennel (sweet), thyme, pennyroyal, cinnamon, camphor, black pepper, cedar wood, organum, nutmeg, peppermint, rose, rosemary
- **Oils not to be used by anybody suffering from high blood pressure:** black pepper, hyssop, rosemary, sage, thyme
- **Oils not to be used by anybody suffering from epilepsy:** camphor, eucalyptus, fennel, hyssop, rosemary, sage, spike lavender (*Lavandula latifolia*), wormwood
- **Oils that make the skin sensitive to ultra-violet light (sunlight and sun beds):** angelica, bergamot, cumin, grapefruit, lemon, lemon verbena, lime, orange, verbena
- **Possibly hazardous and toxic essential oils:** almond, aniseed, cinnamon bark (cinnamon leaf is not hazardous), camphor, clove bud, organum, pine, sage, wintergreen, wormwood

## QUESTIONS:

1. Name the three ancient traditions of herbal medicines, and the names of the ranges of each that we stock at Wellness Warehouse
2. Name a range that we carry that represents Western Herbal Medicine
3. What does the word "Phyto" mean?
4. What is the difference between herbs and spices?
5. Name the 5 main ways we can take or use herbs?
6. Explain in your own words what the difference is between using a whole herb, and a standardised herb
7. What are the 3 categories of interactions with chronic medicines to look out for when recommending herbal medicines to customers?
8. True or False: Customers may stop their prescription chronic medicines and go on to herbal medicines straight away. Why?
9. Name three popular herbs or spices that may have interactions with blood thinning medicines
10. Name three popular herbs or spices that may have interactions with diabetes medication
11. What is the safest approach to the use of herbs during pregnancy?
12. Which herbal teas may be recommended in pregnancy?
13. Which herbal tonics may be recommended in pregnancy?
14. How do adaptogenic herbs support the body?
15. Name 4 popular adaptogenic herbs
16. Name four immune modulating herbs
17. Name four herbs for low mood
18. Name four herbs to assist with urinary tract infections
19. Why should essential oils be diluted in carrier oils before applying them topically?
20. True or False: Essential oils may be taken internally. Why?

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# 7

## MODULE SEVEN

### HOMEOPATHIC MEDICINE

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*"The highest ideal of cure is the speedy, gentle and enduring restoration of health by the most trustworthy and least harmful way"*

- Samuel Hahnemann

# 1. UNDERSTANDING HOMEOPATHY

**History:** In the late 1700's, a German doctor called Samuel Hahnemann first formulated the laws and principles of Homeopathic medicine. He devised an entire system of medicine which he called Homeopathy ("homoios" in Greek means similar, patheia means suffering).

Homeopathic medicine considers the mental, emotional and physical state of the patient. It treats the person and not the disease. Hahnemann developed a vast and meticulously-tested system of Homeopathic medicine and it was the basis for his advancement of the Law of Similars (like cures like).

He found that by increasingly diluting versions of substances taken from all kingdoms of nature (mineral, plant and animal) the effects were enhanced rather than lessened. He began "potentizing" substances by adding water to an original substance and shaking the mixture vigorously between increasing dilutions, to create very potent but gentle and effective medicines.

The number and letter after the name of the remedy refers to how many times the original substance has been shaken and diluted. The most common letter used is C, which stands for "Century" in Latin, meaning 100 – Arnica 30C thus means being diluted 30 x 100. The more diluted and "vibrational" the remedy is the deeper and more powerfully it will work eg 200C for injury.

In 1810, Hahnemann published a text detailing the principles of homeopathic medicine in the first edition of his most important work, Organon of the Healing Art. Hahnemann's work became the foundation for the Homeopathic Materia Medica.

## Simplexes vs Complexes:

Homeopathic remedies are available as single remedies called simplexes, which can only be pre-scribed by homeopathic practitioners or complexes, which are a combination of more than one in-gredient. At Wellness Warehouse we only sell complexes.

## Pillules:

Pillules are small pills, usually in a lactose base. Do not handle the pillules directly as the remedies are very dilute and subtle. Pour 3-5 pillules into the cap of the bottle and then straight into the mouth from the cap. The pillules should be sucked, preferably under the tongue and not chewed or swallowed whole. Each dose is a message to the body rather than an amount of medication, so the dose is the same for an adult, a baby or a horse. Frequency of dosing is the more important aspect.

## Drops:

Homeopathic medicine can also be found in a drop form. These are distilled in an alcohol base. Take the recommended dosage in a little bit of water.

**Homeopathic Brands:** Pegasus, Heel, Boiron, Vitaforce, Natura, Similasan, Rescue Remedy (Bach Flower Remedy)



# 2. USING HOMEOPATHIC REMEDIES

**Absorption of homeopathic remedies** takes place in the mucosa of the mouth. Homeopathic remedies should not be taken within 20 mins of eating or drinking (except water), and **strong flavours** like coffee, peppermint, toothpaste and smoking should be avoided. However, in an emergency situation (e.g. a child has been stung by a bee) give the remedy immediately as some absorption will occur.

The most common dosage regimen e.g. treating a cold or flu, would be: one dose (3 to 5 pillules) every hour for 3 doses, then every 2 hours for 3 doses, thereafter every 4 hours. Once the symptoms are under control, you can reduce to 3 times a day for a few days and then stop.

**The more acute or intense the condition, the more frequent the dose** e.g. in a bee sting, take one dose every 15 minutes for 3 doses, then every half hour for 3 doses. Once there is a positive response, decrease the dosage interval until 4 hourly and stop once resolved.

Until and **infant** can suck the pillules safely without risk of choking (around 8-9 months old), crush 4-5 pillules between 2 plastic medicine measures then add a small amount of water and spoon directly into the child's mouth. Do not use metal spoons as the remedies are very subtle and absorption may be affected. As soon as the **child** can suck the pillules, dose normally.

If you're **dosing frequently** e.g. colic, another method may be useful: dissolve 10 pillules in half a glass of boiled and cooled water and use one medicine measure full for each dose. Discard the water after 24 hours.

Remedies are only indicated for **short-term use (3-5 days)**. As a general rule, if a positive response (symptoms lessening) has not been seen within 24 to 48 hours, advise customers to consult with their homeopathic practitioner.

### Dosage instructions

- Always read the label for the correct dosage instructions
- Do not exceed the dosage as stated on the label

### Contra-indications and interactions

- No side effects
- No interactions with other medicines
- Can't overdose
- Safe to use for the elderly, children and pregnant women



## 3. TISSUE SALTS

These are inorganic minerals that are found in the earth's rocks and soil and makes up our organs, muscles, bone and blood. Tissue salts are needed in our bodies in very small quantities. They are usually ground in lactose into minute particles that are easily absorbed by the human body.

### Dosage instructions

- Use as homeopathic remedies, do not touch the tablets and place the recommended number of tablets under the tongue
- They can also be crushed and given to infants
- Tissue salts can be used over a longer period to help balance the body

### Contra-indications

- Tissue Salt #5 - should not be used with blood thinning medications
- Tissue Salt #12 - eliminates foreign matter from the body, not recommended for persons with metal plates, securing screws, pins, wires, breast implants, pacemakers, contraceptive coils, dental implants and artificial hips, or in pregnancy.

Tissue Salts Brands: Natura, Homeoforce, AllisOne (lactose free, suitable for vegans)



**Tissue Salt # 1:** Calc. Fluor. - Elasticity, flexibility, toning, strength and resilience of muscular and connective tissue, bones, tooth enamel and wall of the blood vessels. Recommend for prolapsed bladder and incontinence

**Tissue Salt # 2:** Calc. Phos. - A cell builder, this is an excellent tonic, as well as growth developer and supporter. It maintains body functions and aids recuperation. It is needed for blood, connective tissue and bones. Recommend as a growth tonic for children or as a recuperation tonic after illness.

**Tissue Salt # 3:** Calc. Sulph. - Natures' cleanser and blood purifier, it dissolves discharge, drains tissues, heals and clears suppuration; an eliminator, it works particularly on the liver, blood and bile.

**Tissue Salt # 4:** Ferrum Phos. - An oxygen transporter; anti-inflammatory; anti-hemorrhage; cooler of inflamed or overheated conditions. Helps the formation of red blood corpuscles and strengthens the blood ves-sels. A first aid remedy. Recommend for inflammation, infection and fever.

**Tissue Salt # 5:** Kali. Mur. - A superb tissue salt for children, specifically for childhood diseases; liver function; decongestant; anti-inflammatory; glandular tonic; blood and lymphatic conditioner; a digestive. Recommend for the second stage of inflammation and sinus infections.

**Tissue Salt # 6:** Kali. Phos. - A nerve nutrient and natural tranquiliser, it lifts the spirits and restores a feeling of well-being; gives emotional balance; pain reliever; important for heart, brain tissue and intra cellular flu-id. Recommend for anxiety and nerve disorders.

**Tissue Salt # 7:** Kali. Sulph. - A cell oxygenator with Ferrum Phos. It transports oxygen; supports liver function; excellent for skin conditions such as eczema; normalises clears and conditions the mucous membranes as it gets rid of mucous. Recommend for dry, scaly skin and scalp, as well as the third stage of inflammation.

**Tissue Salt # 8:** Mag. Phos. - An anti-spasmodic, natural pain reliever for cramps; a superb nerve and muscle relaxant and nutrient. It also treats spasms, tension, bladder stones and stress-related pains and tensions. Recommend for cramps, spasm and headaches.

**Tissue Salt # 9:** Nat. Mur. - For heavy emotions such as anger, depression and irritability; a water distributor for skin conditions, runny nose, hay fever and all mucous membrane conditions. Recommend for all water imbalances and fatigue.

**Tissue Salt # 10:** Nat. Phos. - Nature's antacid and natural acid-alkaline balancer, it treats digestive complaints, arthritic pains and stiffness. It is a mood lifter, acid neutraliser and a stress reliever. Recommend for acidity and rheumatism.

**Tissue Salt # 11:** Nat. Sulph. - Nature's diuretic and toxin cleanser, it is a liver decongestant, and a regulator of body fluid in the whole metabolism. Recommend for nausea and liver disorders.

**Tissue Salt # 12:** Silica - It eliminates toxins from the tissues, clears suppuration, and expels foreign matter from the body. It strengthens connective tissue, supports and sustains after excess stress and overwork, improves memory function and mineral assimilation. Recommend for abscesses, boils and styas.



## QUESTIONS:

1. What are homeopathic complexes?
2. Which homeopathic brands do we sell at Wellness Warehouse?
3. Discuss how homeopathic pillules should be taken
4. Where does the absorption of homeopathic medicines take place?
5. True or False: Homeopathic pillules and tissue salts may be taken straight after brushing the teeth. Why?
6. Choose the correct option: The more acute or intense the condition, the (more/less) frequent the dose. Once the symptoms are under control (increase/reduce) the number of dosages.
7. What are Tissue Salts?
8. Which brand of Tissue Salts is lactose free?
9. Why are homeopathic meds and Tissue Salts such safe medicines to recommend?
10. When would you not recommend Tissue Salt # 5 (Kali. Mur) to a customer?
11. Name 5 instances in which you would not recommend Tissue Salt # 12 (Silica)?
12. Which Tissue Salt would you recommend:
  - For incontinence?
  - As a recuperation tonic after illness?
  - For fever?
  - As a nerve nutrient and natural tranquiliser?
  - For acidic conditions?
  - For cramps and headaches?
  - For water retention?
  - For boils, abscesses and styes?

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## BRINGING IT ALL TOGETHER

**Watch your wording:** we are not allowed to diagnose, prescribe or make claims for medicines:

- "It sounds like you have pneumonia"
- "Use this cream for eczema"
- "This turmeric will cure cancer"

**Your role** is to provide information and different options. It is the **customer's responsibility** to make the correct choice for his/her health and wellness.

- "These are the options that will support your body"
- "These are the differences between the products and the prices. Which product would you like?"

**Ask the right questions to identify the correct products to suggest to your customer:**

- For how long have you experienced these symptoms?
- What caused the onset of these symptoms?
- Are you currently using any natural medicines for this condition? Is it effective?
- What have you used in the past to treat the condition? What was the outcome?
- Are you on any chronic medication?
- Are you on any blood-thinning medications?
- Are you on anti-depressants or anti-anxiety medicines?
- Do you have any allergies?
- Which of these brands do you prefer?

**Factors to take into consideration to match the correct product to your customer:**

- Is the customer a child, a pregnant or lactating woman, elderly or very weakened?
- Do they have any "hot buttons"? Value for money, quality, convenience, vegan/vegetarian, specific diets, allergies and intolerances, cruelty-free or environmentally friendly are all important considerations for customers that we should take into account before recommending products

**When to refer a customer to a qualified practitioner:**

- When they are on chronic medicine - many foods, herbs and supplements might have interactions with chronic medications
- When natural medicines do not have a positive effect on acute conditions within 24 to 48 hours of taking them

**Add On Sales:**

- Think outside the box and give alternatives or add-ons customers may not have thought of, show you care and make suggestions
- Specialised food items, superfoods, teas, essential oils, body care and home care products make nice add on sales – try to cross sell from all three departments (food, health and body care)

**Examples:**

- Have they got the basics covered? Everybody needs a multi-vitamin, a probiotic, an Omega 3 and Vitamin D3.
- Internal medicines to complement topical medicines
- Relaxing body product for someone who is stressed
- Essential oil for inhalations for sinus/head cold
- An air purifier for hay fever



## QUESTIONS:

1. What is the role of the health consultant?
2. What is the responsibility of the customer?
3. Complete the following sentence: We are not allowed to \_\_\_\_\_, prescribe or make \_\_\_\_\_ for medicines.
4. Give 3 examples of important questions to ask customers to identify the correct products to suggest to customers
5. What are typical “hot buttons” that you should listen for when engaging with customers?
6. Name two scenarios when you should refer a customer to a qualified practitioner?
7. Name 3 products which would be suitable to upsell to a customer with digestive complaint to fill his basket?

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